COMPUTER NETWORKS LAB

Course Code: CSE3131

Complete Practical Guide

Aim, Theory, Explanation, Step-by-Step Cisco Packet Tracer Configuration, Testing Procedure, Troubleshooting and Viva Answers for every practical

Department	Computer Science & Engineering
Course	Computer Networks Lab (1 Credit)
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Source	CSE3131 Lab Manual (Experiments 1 - 10 and Additional Experiments 11 - 13)
Tools	Cisco Packet Tracer, NetSim, Nmap, Wireshark

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How to Use This Guide (Read First)

This guide covers every practical in your CSE3131 lab manual. For the experiments that have a Cisco Packet Tracer diagram (Star topology, DHCP, Router/Switch commands, Static Routing, RIP, OSPF, VLAN, NAT and the Case Study) you get: the topology, an addressing table, a cabling table, **every CLI command in the order it must be typed**, how to configure each PC, and a complete **testing / verification** section with the expected output.

1. Pasting commands into the Packet Tracer CLI

- Click the router/switch, open the **CLI** tab, press **Enter** once to get the prompt.
- Lines starting with **!** are comments - IOS ignores them, so you can paste a whole block at once.
- Every config block is written to be pasted **after** you are in privileged mode: type **enable**, then **configure terminal** (the blocks already contain these lines where needed).
- If the router asks *Would you like to enter the initial configuration dialog? [yes/no]* type **no** and press Enter.

2. Cable selection in Packet Tracer (Connections icon, lightning symbol)

Connecting	Cable to choose	Colour / note
PC - Switch, Router - Switch, Server - Switch	Copper Straight-Through	Solid black line
PC - Router, PC - PC, Switch - Switch, Router - Router (Ethernet)	Copper Cross-Over	Dashed black line
Router - Router (WAN)	Serial DCE (click the DCE router first) or Serial DTE	Red zig-zag line; the DCE end needs **clock rate**
Console (optional)	Console cable	Light blue line

TIP: Link lights: red triangle = link down / interface not yet enabled, green = up. After **no shutdown** it can take 20-30 seconds for the lights to turn green. Use the **fast-forward** button (bottom left) to speed this up.

3. How to configure an end device (PC / Server / Laptop)

- Click the PC, open the **Desktop** tab, click **IP Configuration**.
- Choose **Static**, type IP Address, Subnet Mask and Default Gateway (and DNS if needed). Close the window - it saves automatically.
- To test, open **Desktop > Command Prompt** and use **ping**, **tracert**, **ipconfig**, **arp -a**.

4. Always save

After finishing each router/switch: **copy running-config startup-config** (or **write memory**). Save the whole simulation with **File > Save As** (.pkt) - save one file per experiment.

5. Two ways to test in Packet Tracer

- **Realtime mode** (default): use **ping** from the PC Command Prompt.
- **Simulation mode** (button bottom right): click **Add Simple PDU** (closed envelope), click source then destination, then press **Auto Capture / Play** to watch the packet travel hop by hop. Use **Edit Filters** to show only ICMP, ARP, DHCP, RIP, OSPF etc. Open any packet to see the OSI layer details.

List of practicals covered

No.	Practical	Type
1A	Introduction to Packet Tracer and networking device components	Theory + tool walkthrough
1B	Introduction to NetSim and its device components	Theory + tool walkthrough
2	Network utility commands (ping, netstat, ipconfig, ifconfig, arp, tracert, nslookup, pathping)	Commands + practice
3	Star topology using Hub and Switch, IP configuration, show / copy commands, passwords, hostname	Packet Tracer
4	DHCP configuration	Packet Tracer
5	Router modes, switch / router basic commands	Packet Tracer
6	Static routing	Packet Tracer
7	RIPv1 and RIPv2	Packet Tracer
8	OSPF and troubleshooting	Packet Tracer
9	VLAN and troubleshooting	Packet Tracer
10	NAT (static, dynamic, PAT) and troubleshooting	Packet Tracer
11	Nmap, Wireshark, Network scanner	Tools
12	Security threats, attacks, mitigation, device security	Theory + Packet Tracer demo
13	Case study / mini project: build small network and scale it	Packet Tracer

Practical 1A - Introduction to Packet Tracer and Networking Device Components

Aim

To get introduced to Cisco Packet Tracer and to understand the networking device components (routers, switches, end devices, wireless devices, cables) that are used to build networks.

Theory

Packet Tracer is a network simulation tool from Cisco used for learning, teaching and practising network design. It has a drag-and-drop graphical interface, so complete networks can be designed, configured and troubleshot virtually, without any physical hardware. It has two workspaces: **Logical** (build the topology) and **Physical** (rooms, racks, buildings), and two modes: **Realtime** and **Simulation**.

Components of Networking Devices in Packet Tracer

Device	OSI layer	Function	Packet Tracer location / examples
Router	Layer 3 (Network)	Forwards packets between different networks using IP addresses; supports static routing, RIP, OSPF, EIGRP, BGP, NAT, ACL.	Network Devices > Routers (1941, 2911, 2811, Router-PT)
Switch	Layer 2 (Data link)	Connects devices in one LAN, forwards frames using MAC addresses; supports VLAN, STP, trunking.	Network Devices > Switches (2960, Switch-PT)
Hub	Layer 1 (Physical)	Repeats incoming signal to every other port (no intelligence, one collision domain).	Network Devices > Hubs
End devices	Layers 1-7	PCs, laptops, servers, printers, IP phones - source/destination of data. Servers offer HTTP, DNS, DHCP, FTP, Email.	End Devices
Wireless	Layer 1-2	Wireless router / access point creates Wi-Fi (SSID, WPA2).	Network Devices > Wireless Devices
Cables	Layer 1	Copper straight-through, cross-over, fibre, serial, console, coaxial.	Connections

Step-by-step: first practice in Packet Tracer

Step 1: Open Packet Tracer

Start Cisco Packet Tracer. The main window has the workspace in the centre and the **device-type box** at the bottom-left (Network Devices, End Devices, Components, Connections, Misc, Multiuser).

Step 2: Place devices

Click **End Devices** and drag two PCs into the workspace. Click **Network Devices > Switches** and drag one 2960 switch into the workspace.

Step 3: Connect them

Click **Connections**, select **Copper Straight-Through**, click PC0 and choose **FastEthernet0**, then click the switch and choose **FastEthernet0/1**. Repeat for PC1 on **FastEthernet0/2**.

Step 4: Give IP addresses

PC0: 192.168.1.1 / 255.255.255.0, PC1: 192.168.1.2 / 255.255.255.0 (Desktop > IP Configuration).

Step 5: Test

PC0 > Desktop > Command Prompt > **ping 192.168.1.2**. Four replies prove that the basic network works.

Output / Conclusion

The simple network of 2 PCs and 1 switch communicates successfully. This shows the basic use of Packet Tracer for building and testing a network with common components.

Viva Questions with Answers

Q1. What is the role of the switch in the network?

Ans: A switch connects the devices of a LAN and forwards each Ethernet frame only to the port where the destination MAC address is learned (it learns source MACs into a MAC-address table). This reduces collisions and unnecessary traffic.

Q2. Explain the difference between a router and a switch.

Ans: A switch works at Layer 2 with MAC addresses inside one network; a router works at Layer 3 with IP addresses and connects different networks, so it can choose paths (routing), do NAT and apply ACLs. A router separates broadcast domains; a switch (without VLAN) does not.

Q3. Discuss different types of cables and connectors.

Ans: Copper UTP with RJ-45: straight-through (unlike devices, e.g. PC-switch) and cross-over (like devices, e.g. PC-PC, switch-switch); fibre-optic with SC/LC connectors for long-distance and high speed; serial (V.35/DB-60) for WAN links with a DCE/DTE end; console (rollover) cable for device management; coaxial with BNC in old Ethernet / cable TV.

Q4. What is the difference between Bridge and Brouter?

Ans: A bridge connects two LAN segments at Layer 2 and filters using MAC addresses. A brouter (bridge + router) routes routable protocols (like IP) at Layer 3 and bridges the non-routable ones at Layer 2.

Practical 1B - Introduction to NetSim and Device Components

Aim

To study the NetSim network simulation environment and understand the networking device components available for designing and simulating computer networks.

Theory

NetSim (by Tetcos) is a simulation / emulation tool used in teaching and research to design networks, configure devices, analyse protocol behaviour and measure performance (throughput, delay, jitter, packet delivery ratio, packet loss) without physical hardware. It supports Ethernet, Wi-Fi (802.11), MANET, LTE/5G, IoT, sensor, satellite and SDN networks and gives packet-level event traces. In contrast to Packet Tracer (which mimics Cisco IOS and focuses on configuration), NetSim focuses on **performance analysis** using a discrete-event simulation engine.

Components in NetSim

Component	Explanation
Router (Layer 3)	Forwards packets by IP address; static and dynamic routing; many interfaces and routing tables.
Switch (Layer 2)	Forwards Ethernet frames by MAC address; supports VLAN and switching analysis.
End devices (Wired node, server, workstation)	Generate/receive application traffic (HTTP, FTP, Email, Voice, Video).
Wireless access point and mobile nodes	Simulate IEEE 802.11 Wi-Fi; mobile nodes follow mobility models.
Links / media	Ethernet, wireless, fibre, point-to-point, serial - each has bandwidth, delay, error/loss settings.
Internet cloud	Represents the Internet/WAN joining distant networks.
Application / traffic generator	FTP, HTTP, Email, Video, VoIP, custom traffic.
Simulation engine	Discrete-event engine: packet generation, routing, queue management, collision handling, protocol execution, metric calculation.
Performance analyzer	Produces tables/graphs: throughput, end-to-end delay, PDR, loss, jitter, utilisation, queue length, energy.

Procedure (step by step)

Step 1: Launch NetSim and click New Simulation (choose the network type, e.g. Internetworks).

Step 2: From the device palette on the left, drag a Router, a Switch, two Wired Nodes (PCs) and a Server onto the workspace.

Step 3: Connect the devices with Wired Links (click Link icon, click device A then device B).

Step 4: Right-click each device > Properties and set IP address / subnet mask / interface parameters.

Step 5: Click the Application icon, add an application (e.g. FTP or CBR) and choose Source node and Destination node.

Step 6: Click Run Simulation, enter the simulation time (e.g. 10 seconds) and run it. Observe packet animation.

Step 7: When finished NetSim shows the Results Window: open metrics tables, throughput / delay graphs, and the Packet Trace / Event Trace.

Step 8: Save the workspace and note the results for the lab record.

NOTE: NetSim is licensed software available in the college lab. Menu names can differ a little between NetSim versions, but the flow (design - configure - application - run - analyse) is the same.

Output / Result

A basic topology was designed and simulated; traffic flowed between end devices and the performance metrics (throughput, delay, PDR, loss) were generated and analysed.

Viva Questions with Answers

Q1. What is a discrete-event simulator?

Ans: A simulator in which the system state changes only at discrete instants called events (e.g. packet arrival, transmission end), processed from an event queue in time order. NetSim uses this approach.

Q2. Name the main performance metrics shown by NetSim.

Ans: Throughput, end-to-end delay, packet delivery ratio, packet loss, jitter, network utilisation and queue length.

Q3. Difference between Packet Tracer and NetSim?

Ans: Packet Tracer simulates Cisco device configuration (CLI, protocols) for learning; NetSim is a research-grade simulator that gives detailed protocol-level performance statistics and traces.

Practical 2 - Network Utility Commands

Aim

To understand and use the network utility commands PING, NETSTAT, IPCONFIG, IFCONFIG, ARP, TRACEROUTE, NSLOOKUP and PATHPING to monitor and troubleshoot network configuration.

Theory

Network utility commands help to check connectivity, view the IP configuration, see active connections, find the path of packets and resolve names. They are run from the Windows **Command Prompt** or the Linux **Terminal**.

Command reference with explanation

Command	Purpose	Examples	How to read the output
ping	Sends ICMP Echo Request; checks reachability and round-trip time.	ping 192.168.1.1 ping google.com ping -t (Win, continuous) ping -n 10 host	'Reply from' = reachable; time = delay; TTL = remaining hops; 'Request timed out' = no reply (firewall / no route); 'Destination host unreachable' = router has no route.
netstat	Shows active connections, listening ports, routing table, statistics.	netstat -a (all) netstat -n (numeric) netstat -r (route table) netstat -e (Ethernet stats) netstat -b (program, Win admin)	Proto, Local Address:port, Foreign Address:port, State (LISTENING, ESTABLISHED, TIME_WAIT).
ipconfig (Windows)	Shows IP configuration of interfaces.	ipconfig ipconfig /all ipconfig /release ipconfig /renew ipconfig /flushdns	IPv4 address, subnet mask, default gateway; /all adds MAC, DHCP server, DNS.
ifconfig (Linux)	Linux equivalent (newer systems also use **ip addr**).	ifconfig ifconfig eth0 up ip addr show	inet = IP, netmask, broadcast, HWaddr/ether = MAC address, RX/TX packets.
arp	Shows / edits ARP cache (IP-to-MAC mapping).	arp -a arp -d (delete) arp -s IP MAC	Internet Address, Physical Address, Type (dynamic = learned, static = manual).
tracert / traceroute	Shows every router (hop) from source to destination using increasing TTL.	tracert google.com (Win) traceroute google.com (Linux)	Each line = one hop with 3 round-trip times; * * * = hop did not answer.
nslookup	Queries DNS to translate name to IP and back.	nslookup google.com nslookup 8.8.8.8 nslookup -type=mx gmail.com	Server = DNS server used; Name/Address = answer.
pathping (Windows)	Combines ping and tracert; shows packet loss at each hop over time.	pathping google.com	Lost/Sent per hop shows exactly where packets are dropped.

Algorithm / Procedure

Step 1: Open Command Prompt (Windows: Win+R > cmd) or Linux terminal.

Step 2: Type the command with the required parameter (IP or host name).

Step 3: Observe the output.

Step 4: Interpret the result (delay, loss, route, mapping) and note it in the record.

Sample output (as in the manual)

```
C:\>ping 142.250.72.238
Reply from 142.250.72.238: bytes=32 time=25ms TTL=55
Reply from 142.250.72.238: bytes=32 time=26ms TTL=55

C:\>ipconfig
   IPv4 Address. . . . . : 192.168.1.10
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . : 192.168.1.1

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00-14-22-01-23-45     dynamic

C:\>tracert google.com
  1    1 ms    1 ms    1 ms   192.168.1.1
  2   15 ms   15 ms   14 ms   10.100.20.1
  3   25 ms   26 ms   25 ms   142.250.72.238

C:\>nslookup google.com
Server:  dns.google      Address: 8.8.8.8
Name:    google.com      Address: 142.250.72.238
```

Practising these commands inside Packet Tracer

Packet Tracer PCs support **ping, tracert, ipconfig (/all, /renew, /release), arp -a, netstat, nslookup (needs a DNS server), telnet and ssh**. **ifconfig** and **pathping** are not available in the Packet Tracer PC - run them on a real Linux / Windows machine. A good practice is to build the star network of Practical 3 and run:

```
C:\> ipconfig /all
C:\> ping 192.168.1.2
C:\> arp -a
C:\> tracert 192.168.1.2
C:\> netstat
```

TIP: In Packet Tracer the very first ping often shows **Request timed out** once and then 3 replies. This is normal: the first packet is used for ARP resolution.

Conclusion

Different network utility commands were explored. PING checks connectivity, NETSTAT shows connections and ports, IPCONFIG/IFCONFIG display interface settings, ARP shows IP-MAC mapping, TRACEROUTE and PATHPING discover the route and loss per hop, and NSLOOKUP resolves DNS names. These are the first tools of every network administrator.

Viva Questions with Answers

Q1. What is the function of the PING command?

Ans: It sends ICMP Echo Request packets to a host and waits for Echo Reply to test reachability and measure round-trip time and packet loss.

Q2. Explain the difference between IPCONFIG and IFCONFIG.

Ans: IPCONFIG is the Windows command; IFCONFIG is the Linux/Unix command. Both show/configure interface IP settings, but the options and output format differ (ipconfig /all vs ifconfig -a). Modern Linux prefers *ip addr*.

Q3. How does TRACEROUTE help in identifying network issues?

Ans: It sends packets with TTL = 1, 2, 3...; each router that drops a packet returns an ICMP Time-Exceeded message, revealing the router address and delay per hop. The hop where replies stop or delay jumps shows where the problem is.

Q4. What does NSLOOKUP do?

Ans: It queries a DNS server to get the IP address of a domain name (or the name of an IP) and other DNS records like MX, NS.

Practical 3 - Star Topology using HUB and Switch, IP Configuration, Show / Copy Commands, Passwords and Hostname

Aim

To build a star topology with a hub and with a switch, configure IP addresses of end devices, set hostname and passwords on the switch, use **show** commands, save the configuration with the **copy** command and verify communication using ping.

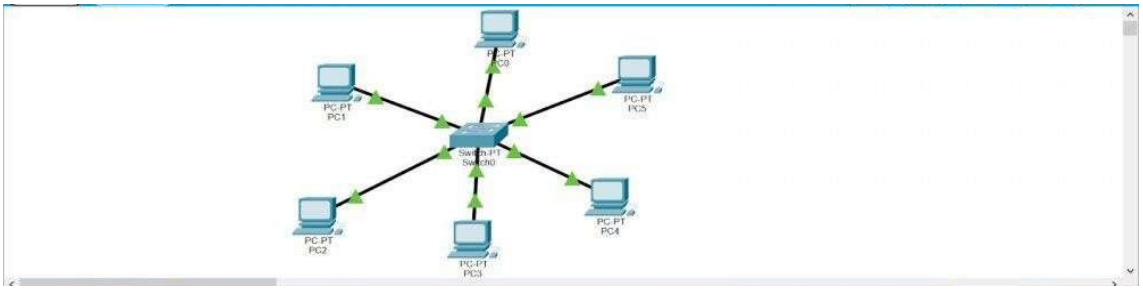
Theory

In a **star topology** every node connects to one central device (hub or switch). All communication passes through the centre. A **hub** is a Layer-1 repeater: whatever enters one port is sent out of all other ports, so all hosts share one collision domain. A **switch** is a Layer-2 device: it learns source MAC addresses and forwards a frame only to the port of the destination, so each port is a separate collision domain. A failure of one cable affects only one node, but failure of the central device stops the whole network.

Requirements

Item	Quantity	Packet Tracer location
PC-PT	6 (PC0 - PC5)	End Devices > PC
Switch 2960 (or Switch-PT)	1	Network Devices > Switches
Generic Hub (for the hub version)	1	Network Devices > Hubs
Copper Straight-Through cables	6	Connections

Topology and addressing



Star topology as given in the lab manual (6 PCs around Switch0)

Device	Interface	Connected to	IP address	Subnet mask
PC0	FastEthernet0	Switch Fa0/1	192.168.1.1	255.255.255.0
PC1	FastEthernet0	Switch Fa0/2	192.168.1.2	255.255.255.0
PC2	FastEthernet0	Switch Fa0/3	192.168.1.3	255.255.255.0
PC3	FastEthernet0	Switch Fa0/4	192.168.1.4	255.255.255.0
PC4	FastEthernet0	Switch Fa0/5	192.168.1.5	255.255.255.0
PC5	FastEthernet0	Switch Fa0/6	192.168.1.6	255.255.255.0

Part A - Star topology using a Switch (step by step)

Step 1: Place the devices

Drag 6 PCs and one 2960 switch to the workspace and arrange the PCs around the switch (as in the diagram). The switch is Switch0 and the PCs are named PC0-PC5.

Step 2: Connect every PC to the switch

Select **Copper Straight-Through**. For each PC click the PC > **FastEthernet0**, then click the switch > the next free port (Fa0/1 ... Fa0/6). After 20-30 seconds all link lights become **green**.

Step 3: Configure the IP address of each PC

PC > Desktop > IP Configuration > Static. Enter the addresses of the addressing table above. Gateway can be left empty because everything is in one network (192.168.1.0/24).

Step 4: Switch configuration: hostname, passwords, banner, save

Click Switch0 > CLI tab and press Enter, then type:

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
SW1(config)# enable secret cisco123            ! encrypted privileged-mode password
SW1(config)# line console 0
SW1(config-line)# password console123          ! console line password
SW1(config-line)# login                         ! ask for it
SW1(config-line)# exit
SW1(config)# line vty 0 15                      ! remote (telnet/ssh) lines
SW1(config-line)# password vty123
SW1(config-line)# login
SW1(config-line)# exit
SW1(config)# service password-encryption        ! encrypt plain-text line passwords
SW1(config)# banner motd #Authorized Access Only#
SW1(config)# interface vlan 1                   ! management IP so the switch can be telnetted
SW1(config-if)# ip address 192.168.1.100 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# end
SW1# copy running-config startup-config         ! press Enter at 'Destination filename'
```

Explanation: *hostname* names the device; *enable secret* protects privileged mode (stored as an MD5 hash); *line console* / *line vty* protect local and remote access; *service password-encryption* hides the plain passwords in the configuration; the *copy running-config startup-config* command copies the RAM configuration into NVRAM so that it survives a reload (the same can be done with *write memory*).

Step 5: Use the show commands

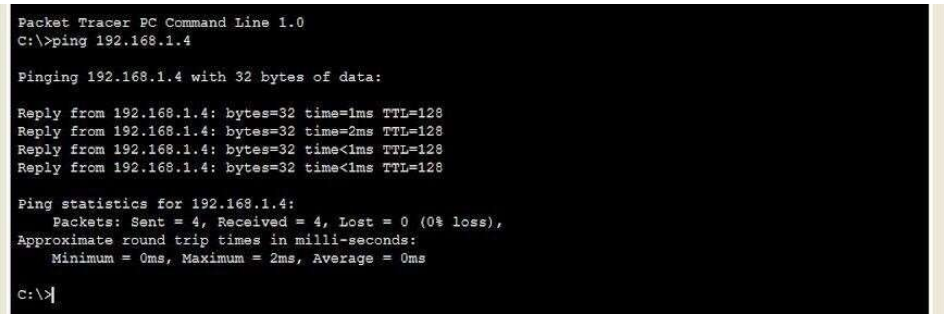
```
SW1# show running-config          ! current configuration (RAM)
SW1# show startup-config          ! saved configuration (NVRAM)
SW1# show version                 ! IOS version, uptime, memory, interfaces
SW1# show interfaces status       ! port status, VLAN, speed, duplex
SW1# show ip interface brief      ! interface IP and up/down state
SW1# show mac address-table       ! MAC addresses learned (after pings)
SW1# show flash                   ! files stored in flash
```

Testing the star topology (switch)

Step 1: Ping test between PCs

PC0 > Desktop > Command Prompt:

```
C:\> ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=2ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```



Expected ping output from the lab manual (PC to 192.168.1.4)

Repeat from other PCs (ping every address 192.168.1.1 - 192.168.1.6). **0% loss** means the topology is correct.

Step 2: Check the MAC address table

After the pings run **show mac address-table** on the switch. You should see 6 dynamic entries, each MAC mapped to Fa0/1 ... Fa0/6 in VLAN 1.

Step 3: Test with PDU in Simulation mode

Switch to **Simulation**, click **Add Simple PDU**, click PC1 and then PC4, press **Auto Capture / Play**. The 'Last Status' column of the PDU list shows **Successful** (as in the manual figure).

Step 4: Test the passwords

- From PC0: Desktop > **Telnet / SSH Client** (or Command Prompt: **telnet 192.168.1.100**). It asks for the vty password (**vty123**); then type **enable** and the secret **cisco123**.
- Run **show running-config** and check that the line passwords are shown encrypted (type 7) and the enable secret as a hash.

Step 5: Verify the saved configuration

Run **show startup-config**. If it prints the configuration, the *copy* command worked. (If 'startup-config is not present', you forgot to save.)

Part B - Star topology using a Hub

Step 1: Replace the switch with a hub

Delete the switch. Place a **Generic Hub** from Network Devices > Hubs, connect the 6 PCs to the hub (Copper Straight-Through, ports Port 0 ... Port 5). Keep the same IP addresses.

Step 2: Test and observe

Ping between PCs - it works the same way. Now use **Simulation mode** with a simple PDU from PC0 to PC3: the **hub copies the packet to all 5 other ports**, while the **switch** (after it has learned the MAC) sends it to the destination port only. This is the key difference to demonstrate in the viva.

Feature	Hub	Switch
OSI layer	1 (Physical)	2 (Data link)
Forwarding	Broadcasts to all ports	Unicast to learned MAC port
Collision domain	One for all ports	One per port
Duplex / speed	Half duplex, shared bandwidth	Full duplex, dedicated bandwidth
Intelligence	None	MAC table, VLAN, security features

Conclusion

A star topology was simulated with both hub and switch. IP addresses were assigned to the PCs and communication was verified using ping. The switch hostname and passwords were set, the configuration was saved with the copy command and the basic show commands were used.

Viva Questions with Answers

Q1. How does a Switch forward packets compared to a HUB?

Ans: A hub repeats every incoming signal to all other ports (flooding). A switch learns source MAC addresses in a MAC table and forwards the frame only to the port of the destination MAC (floods only for unknown/broadcast).

Q2. How can you save the running configuration on a Cisco switch?

Ans: copy running-config startup-config (or **write memory**) in privileged EXEC mode.

Q3. What is the purpose of setting a password on a switch?

Ans: To prevent unauthorised users from accessing the console/remote lines or entering privileged mode and changing the configuration (security).

Q4. How does subnetting help in dividing networks?

Ans: It splits one large network into smaller subnets by borrowing host bits. This reduces broadcast traffic, improves security, uses addresses efficiently and makes management and troubleshooting easier.

Practical 4 - DHCP Configuration

Aim

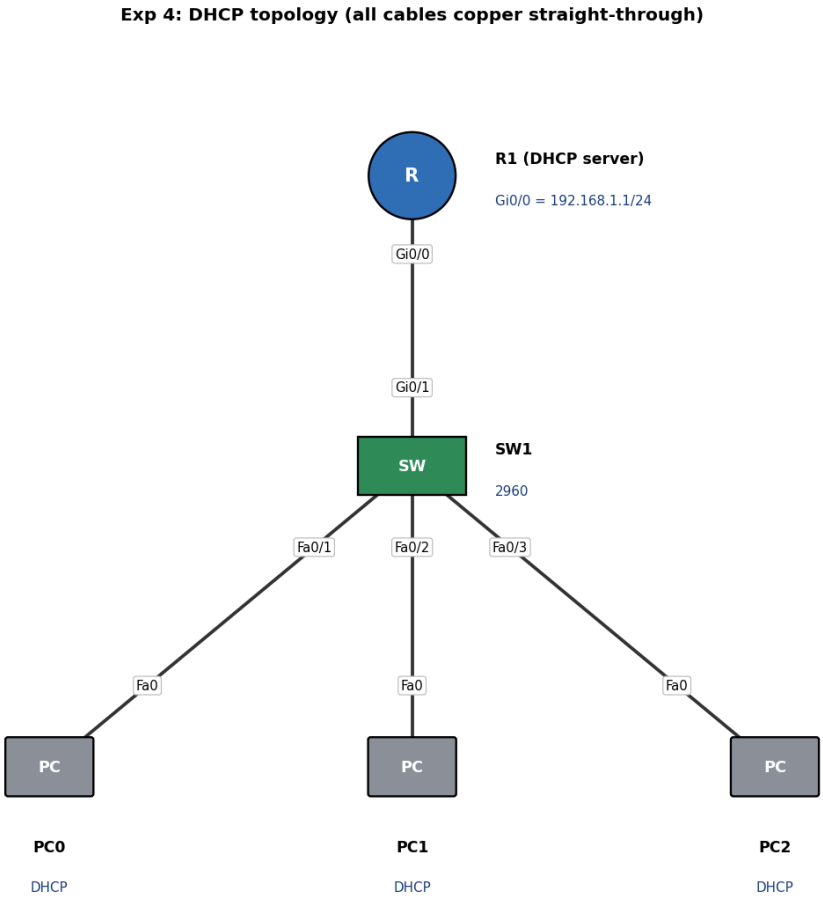
To configure DHCP on a router so that devices in the network get IP addresses automatically, and to verify the configuration.

Theory

DHCP (Dynamic Host Configuration Protocol) automatically gives hosts an IP address, subnet mask, default gateway and DNS server, so nobody has to configure each machine manually and IP conflicts are avoided. It uses UDP port 67 (server) and 68 (client). The process is called **DORA**:

Step	Message	Type	Meaning
1	DHCP Discover	Broadcast (client)	Client looks for a DHCP server.
2	DHCP Offer	Server to client	Server offers an IP address from the pool.
3	DHCP Request	Broadcast (client)	Client requests the offered address.
4	DHCP Acknowledgment	Server to client	Server confirms the lease; the address is bound to the client.

Topology and addressing



DHCP topology - router acts as the DHCP server

Device	Interface	Connected to	Address
R1 (2911)	Gi0/0	SW1 Gi0/1	192.168.1.1 / 255.255.255.0
SW1 (2960)	Fa0/1, Fa0/2, Fa0/3	PC0, PC1, PC2	no IP needed
PC0 - PC2	FastEthernet0	SW1	DHCP (automatic)

DHCP pool design: network 192.168.1.0/24, excluded addresses 192.168.1.1 - 192.168.1.10 (router and reserved devices), default gateway 192.168.1.1, DNS 8.8.8.8. The first PC therefore receives **192.168.1.11**, the second 192.168.1.12 (exactly as in the manual's output).

Step-by-step procedure (router as DHCP server)

Step 1: Build the topology

- Place 1 router (2911), 1 switch (2960) and 3 PCs.
- Router Gi0/0 <-> Switch Gi0/1 using Copper Straight-Through.
- PC0 -> Fa0/1, PC1 -> Fa0/2, PC2 -> Fa0/3 of the switch (Copper Straight-Through).

Step 2: Configure the router interface

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown                      ! interface is shut down by default
R1(config-if)# exit
```

Step 3: Exclude addresses that must not be given out

```
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

This must be typed **before** the pool is used. Addresses 192.168.1.1-.10 are reserved for the router, servers, printers.

Step 4: Create the DHCP pool

```
R1(config)# ip dhcp pool MY_POOL                          ! pool name
R1(dhcp-config)# network 192.168.1.0 255.255.255.0        ! range to lease
R1(dhcp-config)# default-router 192.168.1.1               ! gateway given to clients
R1(dhcp-config)# dns-server 8.8.8.8                       ! DNS given to clients
R1(dhcp-config)# exit
R1(config)# end
R1# copy running-config startup-config
```

Step 5: Configure the PCs as DHCP clients

For each PC: Desktop > IP Configuration > select **DHCP**. The window shows 'DHCP request successful' and fills IP address, mask, default gateway and DNS server.

Testing / verification

Step 1: Check the IP on each PC

```
C:\> ipconfig /all
   IPv4 Address. . . . . : 192.168.1.11
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . : 192.168.1.1
   DNS Servers . . . . . : 8.8.8.8
   DHCP Servers. . . . . : 192.168.1.1
```

Step 2: Check the router

```
R1# show ip dhcp binding
IP address       Client-ID/Hardware address   Lease expiration   Type
192.168.1.11     0001.4A55.A3B7               --                 Automatic
192.168.1.12     0002.4B67.FAD9               --                 Automatic

R1# show ip dhcp pool
Pool MY_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Pool MY_POOL : 2 addresses in use, 246 available   (as shown in the manual)
```

```
R1# show ip dhcp conflict         ! should be empty
R1# show running-config | include dhcp
```

(MAC addresses will differ in your simulation. The numbers 'in use / available' will change with the number of PCs.)

Step 3: Test connectivity

PC0: **ping 192.168.1.1** (gateway) and **ping 192.168.1.12** (another PC) - both must reply.

Step 4: Watch DORA in Simulation mode

Switch to Simulation, **Edit Filters > show only DHCP**. On PC2 go to IP Configuration, select Static and then DHCP again. Press Play and watch Discover, Offer, Request and ACK; click each envelope to see the fields (Offered IP, lease, gateway).

Step 5: Renew test

```
C:\> ipconfig /release
C:\> ipconfig /renew
```

Alternative: a Server as DHCP server

Place a Server-PT on the switch, give it the static address 192.168.1.2/24. Server > **Services** tab > **DHCP**: Service = **On**; Pool Name = MY_POOL; Default Gateway = 192.168.1.1; DNS Server = 8.8.8.8; Start IP Address = 192.168.1.11; Subnet Mask = 255.255.255.0; Maximum Users = 100; click **Save**. Set the PCs to DHCP and verify the same way.

Troubleshooting

Problem	Reason	Fix
PC shows 169.254.x.x (APIPA)	No DHCP server reachable / interface down	Check cable lights, **no shutdown** on Gi0/0, pool network matches the interface network
PC gets address of excluded range	Excluded-address command missing	Add **ip dhcp excluded-address** and renew
No default gateway on PC	**default-router** command missing	Add it inside the pool
Pool exhausted	More clients than addresses	Increase range, reduce lease or exclude fewer addresses

Conclusion

DHCP was configured on a router; PCs received IP address, mask, gateway and DNS automatically. DHCP simplifies administration and avoids duplicate addresses, but it must be secured (e.g. DHCP snooping, limited pools) against rogue DHCP servers and address exhaustion.

Viva Questions with Answers

Q1. What is the role of a DHCP server in a network?

Ans: It automatically assigns IP addresses and related settings (mask, gateway, DNS) to clients from a pool and manages leases, preventing conflicts and manual configuration.

Q2. Explain the DHCP Discover and DHCP Offer processes.

Ans: Discover: a new client broadcasts (src 0.0.0.0, dst 255.255.255.255, UDP 68->67) to find a server. Offer: the server replies with an available IP address, mask, lease time and other options.

Q3. What is the significance of the default-router command in DHCP configuration?

Ans: It tells clients which gateway address to use to reach other networks; without it the PCs get an IP address but cannot leave their subnet.

Q4. Why would you use the ip dhcp excluded-address command?

Ans: To keep some addresses (router interface, servers, printers) out of the pool so they are never leased to clients, avoiding IP conflicts.

Practical 5 - Router Modes, Switch / Router Basic Commands

Aim

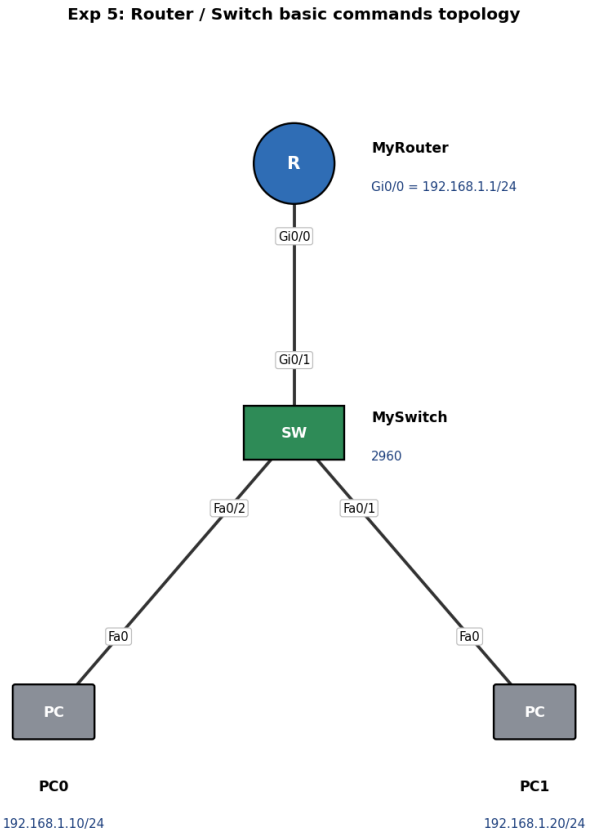
To understand the modes of operation of a Cisco router/switch and to practise the basic commands for configuring and managing them.

Theory - command modes

Mode	Prompt	How to enter	What you can do
User EXEC	Router>	Default when you log in	Limited, view basic info (ping, show version)
Privileged EXEC	Router#	**enable**	All show commands, debug, copy, reload
Global configuration	Router(config)#	**configure terminal**	System-wide settings (hostname, passwords, routing, VLANs)
Interface configuration	Router(config-if)#	**interface gigabitEthernet 0/0**	IP address, no shutdown, description
Line configuration	Router(config-line)#	**line console 0** / **line vty 0 15**	Console / remote access passwords
Router configuration	Router(config-router)#	**router rip** / **router ospf 1**	Routing protocol settings
VLAN configuration (switch)	Switch(config-vlan)#	**vlan 10**	VLAN name

Moving between modes: *enable* (user to privileged), *disable* (back), *configure terminal* (to global), *exit* (one level back), *end* or **Ctrl+Z** (straight back to privileged). The **?** key shows the available commands, **Tab** completes a command and commands can be shortened (conf t, sh run, int gi0/0).

Topology and addressing



Topology for Practical 5

Device	Interface	Connected to	Address
MyRouter (2911)	Gi0/0	MySwitch Gi0/1	192.168.1.1 / 255.255.255.0
MySwitch (2960)	Fa0/1	PC1	VLAN 10 (Sales)
MySwitch	Fa0/2	PC0	VLAN 1 (default)
PC0	Fa0	Fa0/2	192.168.1.10 / 255.255.255.0, gw 192.168.1.1
PC1	Fa0	Fa0/1	192.168.1.20 / 255.255.255.0, gw 192.168.1.1

Step-by-step

Step 1: Build the network

Place router, switch, 2 PCs. Router Gi0/0 to Switch Gi0/1, PC1 to Fa0/1, PC0 to Fa0/2 (all straight-through). Set the PC IP addresses as in the table.

Step 2: Explore the modes on the router

```
Router> ?                          ! list of user-mode commands
Router> enable                     ! -> Router#
Router# configure terminal         ! -> Router(config)#
Router(config)# interface gigabitEthernet 0/0     ! -> Router(config-if)#
Router(config-if)# exit            ! back to global
Router(config)# line console 0     ! -> Router(config-line)#
Router(config-line)# end           ! straight back to Router#
Router# disable                    ! back to Router>
```

Step 3: Basic router configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname MyRouter
MyRouter(config)# enable secret cisco123
MyRouter(config)# banner motd #Authorized Access Only#
MyRouter(config)# line console 0
MyRouter(config-line)# password console123
MyRouter(config-line)# login
MyRouter(config-line)# exit
MyRouter(config)# service password-encryption
MyRouter(config)# interface gigabitEthernet 0/0
MyRouter(config-if)# description LAN to MySwitch
MyRouter(config-if)# ip address 192.168.1.1 255.255.255.0
MyRouter(config-if)# no shutdown               ! bring the interface up
MyRouter(config-if)# end
MyRouter# copy running-config startup-config
```

Why no shutdown? Router interfaces are administratively down by default. The command turns the interface on; without it the line protocol stays down and nothing can pass.

Step 4: Basic switch configuration including VLAN

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname MySwitch
MySwitch(config)# enable secret cisco123
MySwitch(config)# vlan 10
MySwitch(config-vlan)# name Sales
MySwitch(config-vlan)# exit
MySwitch(config)# interface fastEthernet 0/1
MySwitch(config-if)# switchport mode access
MySwitch(config-if)# switchport access vlan 10
MySwitch(config-if)# exit
MySwitch(config)# interface vlan 1
MySwitch(config-if)# ip address 192.168.1.2 255.255.255.0
MySwitch(config-if)# no shutdown
MySwitch(config-if)# end
MySwitch# copy running-config startup-config
```

Testing / verification

```
MyRouter# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down

MyRouter# show running-config          ! full configuration
MyRouter# show startup-config
MyRouter# show version                 ! IOS version, uptime, RAM, flash
MyRouter# show interfaces gigabitEthernet 0/0
MyRouter# show ip route                ! 'C 192.168.1.0/24 is directly connected'
MyRouter# show history                 ! last typed commands

MySwitch# show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, ... Gig0/2
10   Sales                            active    Fa0/1
MySwitch# show interfaces fa0/1 switchport
MySwitch# show mac address-table
```

- **PC0 > ping 192.168.1.1** must succeed (PC0 is in VLAN 1 like the router port).
- **PC1 > ping 192.168.1.1** will **fail** because PC1 is in VLAN 10 and the router is in VLAN 1. This proves that VLANs separate traffic. Move PC1 back (*switchport access vlan 1*) and the ping works.
- Close the CLI, reopen it and type **enable**: it must ask for the password **cisco123**.
- Type **reload** after saving: the hostname and IP persist. Without saving they would be lost.

Useful extra commands

Command	Use
do show running-config	Run an EXEC command from inside a configuration mode
show ip interface brief	Quick summary of all interface IPs and status
erase startup-config; reload	Factory reset the device
ip default-gateway 192.168.1.1 (switch)	Default gateway of a switch for management
no ip domain-lookup	Stop the router from trying to DNS-resolve mistyped commands
show clock / clock set	Show / set time

Conclusion

The command modes of Cisco devices were studied. The router hostname, password, banner and interface IP were configured, VLAN was created on the switch and verified using the show commands.

Viva Questions with Answers

Q1. What is the difference between User EXEC Mode and Privileged EXEC Mode?

Ans: User EXEC (Router>) gives limited, mostly read-only commands; Privileged EXEC (Router#) gives full access to show, debug, copy, reload and is the doorway to configuration mode.

Q2. Why do we use the no shutdown command after configuring an interface?

Ans: Router interfaces are administratively shut down by default; *no shutdown* enables them so the interface can come up.

Q3. What is the purpose of setting a default gateway on a router/host?

Ans: It is the next hop used to reach destinations outside the local network. On a host it is the router interface; on a router a default route (0.0.0.0/0) sends all unknown traffic to the next router.

Practical 6 - Static Routing

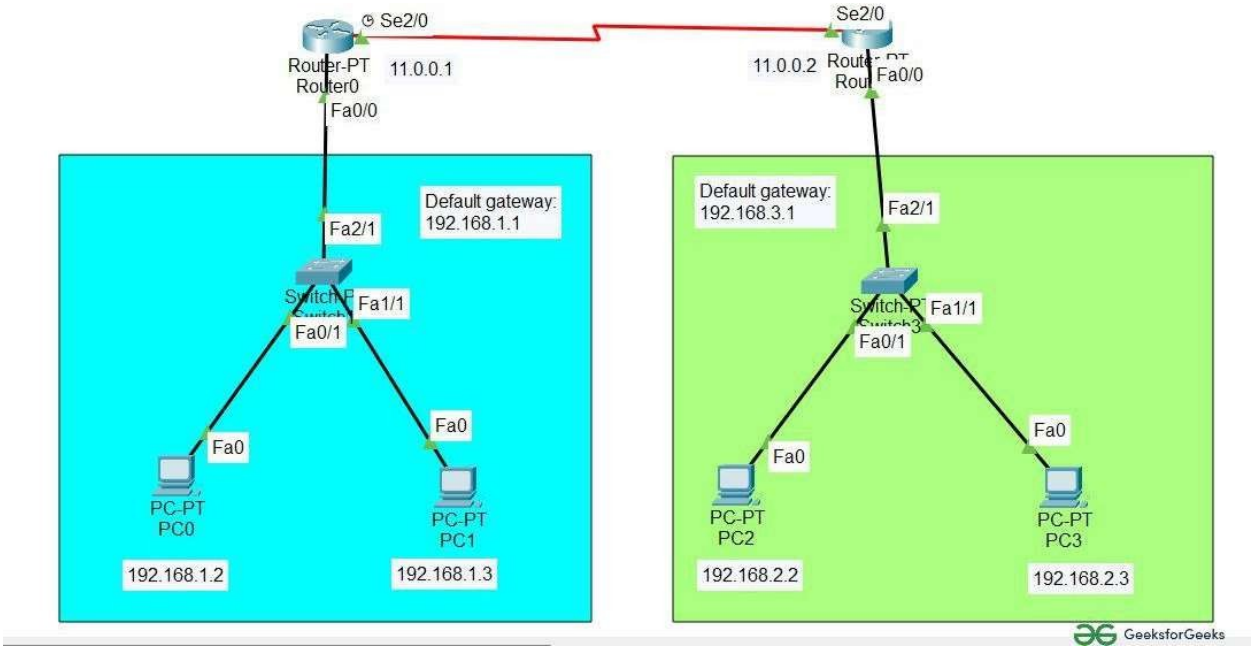
Aim

To configure static routing between two routers so that PCs of two different networks can communicate through manually defined routes.

Theory

In static routing the administrator manually writes the routes into the routing table. Static routes do not change automatically when the topology changes, use no CPU/bandwidth for updates, are predictable and secure, and are ideal for small networks and stub networks. Each router automatically knows only its **directly connected** networks; it must be told how to reach every **remote** network with: **ip route <destination network> <mask> <next-hop IP or exit interface>**. The default administrative distance of a static route is **1** (connected = 0, RIP = 120, OSPF = 110).

Topology



Static routing topology from the lab manual (two LANs joined by a serial link)

IMPORTANT: The manual's figure labels the right-hand default gateway as 192.168.3.1 although the PCs are 192.168.2.2 and 192.168.2.3. To make the network consistent this guide uses **192.168.2.0/24** for the right LAN, so the gateway of PC2/PC3 is **192.168.2.1**.

Requirements, addressing and cabling

Device	Interface	IP address	Mask	Gateway
Router0	Fa0/0	192.168.1.1	255.255.255.0	-
Router0	Se2/0 (DCE)	11.0.0.1	255.0.0.0	-
Router1	Fa0/0	192.168.2.1	255.255.255.0	-
Router1	Se2/0 (DTE)	11.0.0.2	255.0.0.0	-
PC0	Fa0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	Fa0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	Fa0	192.168.2.3	255.255.255.0	192.168.2.1

From	To	Cable
PC0 Fa0	Switch (left) Fa0/1	Copper Straight-Through
PC1 Fa0	Switch (left) Fa1/1	Copper Straight-Through
Switch (left) Fa2/1	Router0 Fa0/0	Copper Straight-Through
PC2 Fa0	Switch (right) Fa0/1	Copper Straight-Through
PC3 Fa0	Switch (right) Fa1/1	Copper Straight-Through
Switch (right) Fa2/1	Router1 Fa0/0	Copper Straight-Through
Router0 Se2/0	Router1 Se2/0	Serial DCE (click Router0 first so Router0 is the DCE end)

NOTE: The figure uses **Router-PT** and **Switch-PT**. The serial port Se2/0 appears after you **switch the router off** (click the power switch in the Physical tab), drag module **PT-ROUTER-NM-1S** into the free slot, then switch it on again. If your Packet Tracer does not have Router-PT, use 1841/2811 routers with a WIC-2T serial card and replace *Se2/0* with the interface name of your card (e.g. *Se0/0/0*) in all commands.

Step-by-step

Step 1: Build the topology

Place 2 routers, 2 switches, 4 PCs. Add the serial module to both routers. Connect the devices as in the cabling table.

Step 2: Configure the PCs

Desktop > IP Configuration > Static: use the values in the addressing table (do not forget the **Default Gateway**).

Step 3: Configure Router0

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface fastEthernet 0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface serial 2/0
Router0(config-if)# ip address 11.0.0.1 255.0.0.0
Router0(config-if)# clock rate 64000          ! ONLY on the DCE side (clock icon on the cable)
Router0(config-if)# no shutdown
Router0(config-if)# exit
!  --- STATIC ROUTE: to reach 192.168.2.0/24 send packets to next-hop 11.0.0.2 ---
Router0(config)# ip route 192.168.2.0 255.255.255.0 11.0.0.2
Router0(config)# end
Router0# copy running-config startup-config
```

Step 4: Configure Router1

```
Router> enable
Router# configure terminal
Router(config)# hostname Router1
Router1(config)# interface fastEthernet 0/0
Router1(config-if)# ip address 192.168.2.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface serial 2/0
Router1(config-if)# ip address 11.0.0.2 255.0.0.0
Router1(config-if)# no shutdown                ! no clock rate on the DTE side
Router1(config-if)# exit
!  --- STATIC ROUTE: to reach 192.168.1.0/24 send packets to next-hop 11.0.0.1 ---
Router1(config)# ip route 192.168.1.0 255.255.255.0 11.0.0.1
Router1(config)# end
Router1# copy running-config startup-config
```

Explanation of the route command: *ip route 192.168.2.0 255.255.255.0 11.0.0.2* means 'any packet for network 192.168.2.0/24 must be sent to the neighbour 11.0.0.2'. The return route on Router1 is needed too - without it the

ping request arrives but the reply cannot return.

Optional variants (for the viva)

```
Router0(config)# ip route 192.168.2.0 255.255.255.0 serial 2/0     ! exit-interface form
Router1(config)# ip route 0.0.0.0 0.0.0.0 11.0.0.1                  ! default route (stub router)
Router0(config)# ip route 192.168.2.0 255.255.255.0 11.0.0.2 150    ! floating static route (AD 150)
```

Testing / verification

Step 1: Check interfaces and routing tables

```
Router0# show ip interface brief
FastEthernet0/0   192.168.1.1   YES manual up   up
Serial2/0         11.0.0.1      YES manual up   up

Router0# show ip route
C    11.0.0.0/8 is directly connected, Serial2/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 11.0.0.2

Router1# show ip route
C    11.0.0.0/8 is directly connected, Serial2/0
S    192.168.1.0/24 [1/0] via 11.0.0.1
C    192.168.2.0/24 is directly connected, FastEthernet0/0

Router0# show ip route static          ! lists only the S routes
Router0# show running-config | include ip route
```

C = connected, **S** = static, **[1/0]** = administrative distance 1 / metric 0.

Step 2: Ping test

```
PC0> ping 192.168.1.1      (own gateway - must work)
PC0> ping 11.0.0.1         (router0 serial)
PC0> ping 11.0.0.2         (router1 serial - tests the WAN link)
PC0> ping 192.168.2.2      (PC2 on the other LAN - tests static routing)
PC0> ping 192.168.2.3
PC2> ping 192.168.1.2      (reverse direction)
```

Step 3: Trace the path

```
PC0> tracert 192.168.2.2
  1   1 ms   0 ms   0 ms   192.168.1.1     (Router0 Fa0/0)
  2   2 ms   1 ms   1 ms   11.0.0.2        (Router1 Se2/0)
  3   3 ms   2 ms   2 ms   192.168.2.2     (PC2)
```

Step 4: Simulation mode

Add a simple PDU from PC0 to PC3 and press Play. The packet moves PC0 - Switch - Router0 - Router1 - Switch - PC3 and returns; the status becomes **Successful**.

Troubleshooting

Symptom	Likely cause	How to check / fix
Serial link is down (red)	Missing *clock rate* on DCE, or interface shut	*show controllers serial 2/0* (shows DCE/DTE); add *clock rate 64000*; *no shutdown*
'Destination host unreachable' from 192.168.1.1	Router0 has no route to the destination	*show ip route*, add the static route
'Request timed out'	Return route missing on the far router, or wrong gateway on a PC	Check *ip route* on both routers and PC gateway
Route appears but ping fails	Wrong next-hop address or mask	Compare next-hop with neighbour's serial IP
Route not in table	Next-hop not reachable (interface down)	A static route is installed only when the next-hop/exit interface is up

Conclusion

Static routing was configured between two routers. By manually adding the routes, devices in the two different networks could communicate; this was proven with ping and tracert and the S routes in the routing tables.

Viva Questions with Answers

Q1. What is the difference between static routing and dynamic routing?

Ans: Static routes are entered manually and do not adapt to failures; dynamic routing protocols (RIP, OSPF, EIGRP) exchange information and automatically learn and update routes.

Q2. What are the advantages and disadvantages of static routing?

Ans: Advantages: no protocol overhead, simple, secure, predictable. Disadvantages: manual work, not scalable, no automatic recovery from link failure, error-prone in large networks.

Q3. Explain the next-hop IP address in static routing.

Ans: It is the IP address of the directly connected neighbouring router interface to which packets for the destination network must be forwarded.

Q4. How can you verify that a static route has been correctly configured on a router?

Ans: *show ip route* (look for the 'S' entry), *show running-config* | *include ip route*, then ping and traceroute to the remote network.

Practical 7 - RIPv1 and RIPv2

Aim

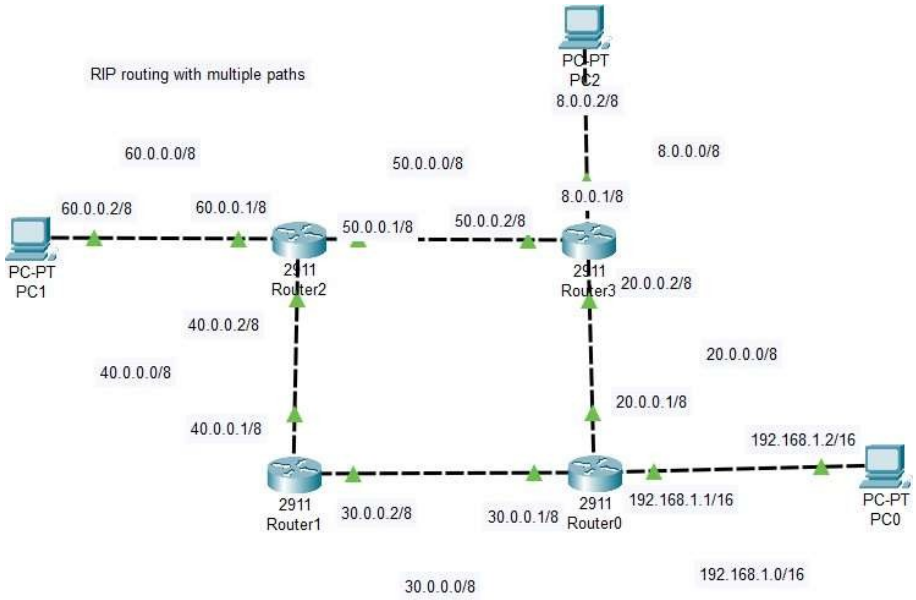
To configure the Routing Information Protocol (RIP) version 1 and version 2 on a set of routers so that routes are learned dynamically between several networks (including multiple paths).

Theory

RIP is a **distance-vector** interior routing protocol. Metric = **hop count**; the maximum is **15** hops (16 means unreachable). Routers send their full routing table every **30 seconds** (update timer); invalid timer 180 s, flush timer 240 s. Administrative distance = **120**. Loop prevention uses split horizon, route poisoning and hold-down.

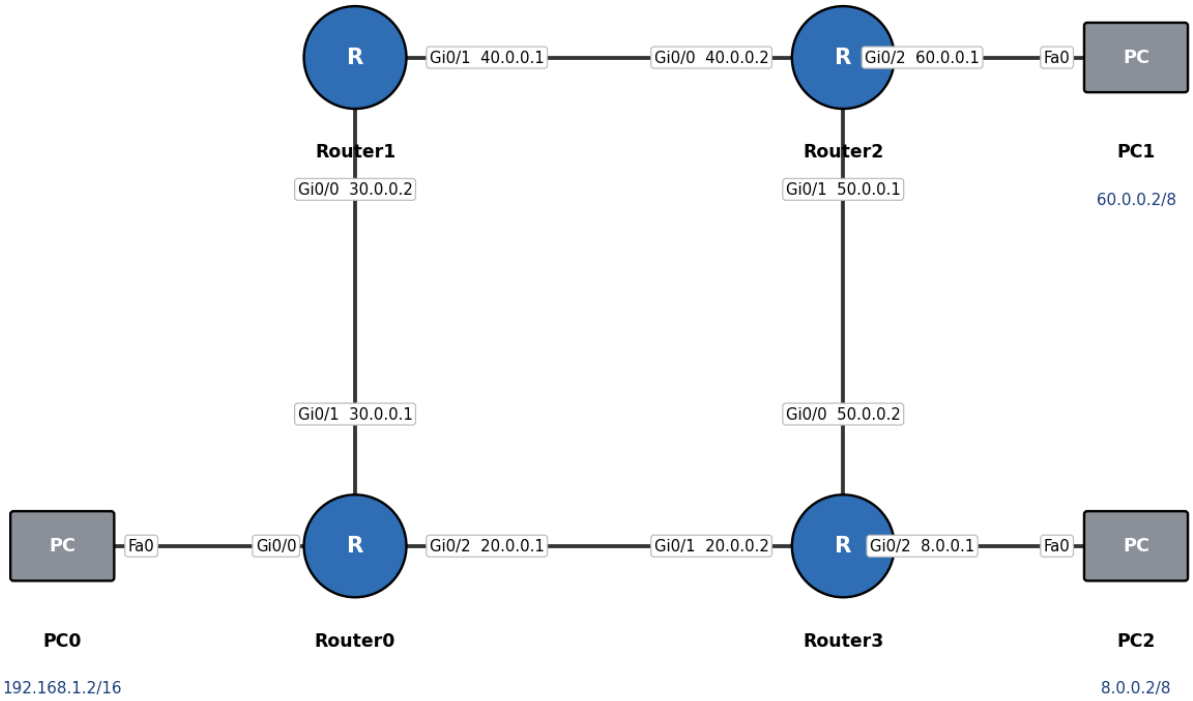
Feature	RIPv1	RIPv2
Type	Classful (no subnet mask in updates)	Classless (mask sent in updates)
VLSM / CIDR	Not supported	Supported
Update method	Broadcast 255.255.255.255	Multicast 224.0.0.9
Authentication	No	Yes (plain / MD5)
Auto-summary	Always at classful boundary	On by default; disable with *no auto-summary*

Topology



RIP topology from the manual (four 2911 routers, multiple paths)

Exp 7: RIP topology with port numbers (copper cross-over cables everywhere)



Same topology with interface numbers used in this guide

Addressing table

Device	Interface	IP address	Mask	Connected to
Router0	Gi0/0	192.168.1.1	255.255.0.0 (/16)	PC0
Router0	Gi0/1	30.0.0.1	255.0.0.0	Router1 Gi0/0
Router0	Gi0/2	20.0.0.1	255.0.0.0	Router3 Gi0/1
Router1	Gi0/0	30.0.0.2	255.0.0.0	Router0 Gi0/1
Router1	Gi0/1	40.0.0.1	255.0.0.0	Router2 Gi0/0
Router2	Gi0/0	40.0.0.2	255.0.0.0	Router1 Gi0/1
Router2	Gi0/1	50.0.0.1	255.0.0.0	Router3 Gi0/0
Router2	Gi0/2	60.0.0.1	255.0.0.0	PC1
Router3	Gi0/0	50.0.0.2	255.0.0.0	Router2 Gi0/1
Router3	Gi0/1	20.0.0.2	255.0.0.0	Router0 Gi0/2
Router3	Gi0/2	8.0.0.1	255.0.0.0	PC2
PC0	Fa0	192.168.1.2	255.255.0.0	gw 192.168.1.1
PC1	Fa0	60.0.0.2	255.0.0.0	gw 60.0.0.1
PC2	Fa0	8.0.0.2	255.0.0.0	gw 8.0.0.1

NOTE: All links in this figure are **Copper Cross-Over** (router to router and router to PC). The manual uses a /16 mask on 192.168.1.0; that is kept here. For RIPv1 the *network* command still takes the classful network (192.168.1.0).

Step-by-step

Step 1: Build the topology and set the PCs

Place four 2911 routers and three PCs, connect as in the table, and set the PC IP addresses and gateways.

Step 2: Configure the interfaces of every router (common part)

Router0

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface gigabitEthernet 0/0
Router0(config-if)# ip address 192.168.1.1 255.255.0.0
Router0(config-if)# no shutdown
Router0(config-if)# interface gigabitEthernet 0/1
Router0(config-if)# ip address 30.0.0.1 255.0.0.0
Router0(config-if)# no shutdown
Router0(config-if)# interface gigabitEthernet 0/2
Router0(config-if)# ip address 20.0.0.1 255.0.0.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
```

Router1

```
Router> enable
Router# configure terminal
Router(config)# hostname Router1
Router1(config)# interface gigabitEthernet 0/0
Router1(config-if)# ip address 30.0.0.2 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# interface gigabitEthernet 0/1
Router1(config-if)# ip address 40.0.0.1 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

Router2

```
Router> enable
Router# configure terminal
Router(config)# hostname Router2
Router2(config)# interface gigabitEthernet 0/0
Router2(config-if)# ip address 40.0.0.2 255.0.0.0
Router2(config-if)# no shutdown
Router2(config-if)# interface gigabitEthernet 0/1
Router2(config-if)# ip address 50.0.0.1 255.0.0.0
Router2(config-if)# no shutdown
Router2(config-if)# interface gigabitEthernet 0/2
Router2(config-if)# ip address 60.0.0.1 255.0.0.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
```

Router3

```
Router> enable
Router# configure terminal
Router(config)# hostname Router3
Router3(config)# interface gigabitEthernet 0/0
Router3(config-if)# ip address 50.0.0.2 255.0.0.0
Router3(config-if)# no shutdown
Router3(config-if)# interface gigabitEthernet 0/1
Router3(config-if)# ip address 20.0.0.2 255.0.0.0
Router3(config-if)# no shutdown
Router3(config-if)# interface gigabitEthernet 0/2
Router3(config-if)# ip address 8.0.0.1 255.0.0.0
Router3(config-if)# no shutdown
Router3(config-if)# exit
```

Step 3: Enable RIP version 1 on every router and advertise the connected networks

Each router advertises only its **directly connected** networks:

```
! ---------- Router0 ----------
Router0(config)# router rip
Router0(config-router)# version 1
Router0(config-router)# network 192.168.1.0
Router0(config-router)# network 30.0.0.0
Router0(config-router)# network 20.0.0.0
Router0(config-router)# end
Router0# copy running-config startup-config

! ---------- Router1 ----------
Router1(config)# router rip
Router1(config-router)# version 1
Router1(config-router)# network 30.0.0.0
Router1(config-router)# network 40.0.0.0
Router1(config-router)# end
Router1# copy running-config startup-config

! ---------- Router2 ----------
Router2(config)# router rip
Router2(config-router)# version 1
Router2(config-router)# network 40.0.0.0
Router2(config-router)# network 50.0.0.0
Router2(config-router)# network 60.0.0.0
Router2(config-router)# end
Router2# copy running-config startup-config

! ---------- Router3 ----------
Router3(config)# router rip
Router3(config-router)# version 1
Router3(config-router)# network 50.0.0.0
Router3(config-router)# network 20.0.0.0
Router3(config-router)# network 8.0.0.0
Router3(config-router)# end
Router3# copy running-config startup-config
```

Step 4: Convert to RIPv2 (do this on all four routers)

```
Router0# configure terminal
Router0(config)# router rip
Router0(config-router)# version 2
Router0(config-router)# no auto-summary        ! send real masks (needed for /16 and VLSM)
Router0(config-router)# end
! repeat the same three lines (router rip / version 2 / no auto-summary) on Router1, Router2 and Router3
```

Testing / verification

Step 1: Verify RIP routes (wait 30-60 s or use fast-forward)

```
Router0# show ip route
C    20.0.0.0/8 is directly connected, GigabitEthernet0/2
C    30.0.0.0/8 is directly connected, GigabitEthernet0/1
C    192.168.0.0/16 is directly connected, GigabitEthernet0/0
R    8.0.0.0/8  [120/1] via 20.0.0.2, 00:00:12, GigabitEthernet0/2
R    40.0.0.0/8 [120/1] via 30.0.0.2, 00:00:20, GigabitEthernet0/1
R    50.0.0.0/8 [120/1] via 20.0.0.2, 00:00:12, GigabitEthernet0/2
R    60.0.0.0/8 [120/2] via 30.0.0.2, 00:00:20, GigabitEthernet0/1
                [120/2] via 20.0.0.2, 00:00:12, GigabitEthernet0/2     <- two equal-cost paths
```

R = learned by RIP, **[120/2]** = administrative distance 120, metric 2 hops. Network 60.0.0.0 shows two paths of equal metric (Router0-Router1-Router2 and Router0-Router3-Router2) - RIP load-balances over them.

```
Router0# show ip protocols             ! RIP version, timers, networks, neighbours
Router0# show ip rip database          ! all RIP routes with metrics
Router0# debug ip rip                  ! live updates (stop with: undebug all)
```

Step 2: End-to-end tests

```
PC0> ping 60.0.0.2        (PC0 to PC1)
PC0> ping 8.0.0.2         (PC0 to PC2)
PC1> ping 8.0.0.2
PC0> tracert 8.0.0.2      (should go PC0 - Router0 - Router3 - PC2)
```

Step 3: Test convergence / alternate path

```
Router0# configure terminal
Router0(config)# interface gigabitEthernet 0/2
Router0(config-if)# shutdown               ! cut the direct Router0-Router3 link
Router0(config-if)# end
```

Wait about 30-60 seconds and run **show ip route** again: network 8.0.0.0 is now reached with metric 3 via 30.0.0.2 (Router1-Router2-Router3). **tracert 8.0.0.2** from PC0 now shows the longer path and the ping still works. Then run **no shutdown** on Gi0/2 and the shorter path returns. This demonstrates dynamic routing.

Step 4: Compare RIPv1 and RIPv2

- With **version 1**, *show ip route* on Router1 shows the remote LAN as **192.168.1.0/24** (classful guess).
- With **version 2 + no auto-summary** it shows **192.168.0.0/16** - the real mask from Router0.
- In Simulation mode with the RIP filter: v1 packets are broadcast, v2 packets are multicast to 224.0.0.9.

Troubleshooting

Problem	Cause	Fix
No R routes	Interface down / wrong network statement	*show ip interface brief*; *show ip protocols* lists the advertised networks
Routes missing after mixing versions	Some routers v1, others v2	Use the same version on every router
Discontiguous networks / wrong summary	Auto-summary	*no auto-summary* with RIPv2
Destination 16 hops away	RIP limit is 15	Redesign or use OSPF

Conclusion

RIPv1 and RIPv2 were configured on four routers. Routes were propagated dynamically, multiple paths and automatic recovery from a link failure were demonstrated. RIPv2 is classless and therefore better for modern networks.

Viva Questions with Answers

Q1. What is the maximum hop count allowed by RIP?

Ans: 15 hops; a metric of 16 means the network is unreachable.

Q2. Explain the differences between RIPv1 and RIPv2.

Ans: RIPv1: classful, broadcast updates, no authentication, no VLSM. RIPv2: classless (sends subnet mask), multicast 224.0.0.9, supports authentication, VLSM and CIDR.

Q3. Why is RIPv1 considered a classful routing protocol?

Ans: Its updates carry no subnet mask, so receivers assume the default class A/B/C mask (or the mask of the receiving interface) and networks are summarised at classful boundaries.

Q4. What are the limitations of RIPv1 in modern networks?

Ans: No VLSM/CIDR, no discontiguous subnets, broadcast updates waste bandwidth, no authentication, 15-hop limit, slow convergence and hop-count-only metric.

Practical 8 - OSPF Configuration and Troubleshooting

Aim

To configure the OSPF routing protocol on routers, verify neighbour relationships and routes, and troubleshoot common OSPF problems.

Theory

OSPF (Open Shortest Path First) is a **link-state** interior gateway protocol. Each router floods **LSAs** describing its links, builds the same link-state database, and runs Dijkstra's SPF algorithm to find the shortest path. Metric = **cost** = reference bandwidth (100 Mbps) / interface bandwidth (FastEthernet = 1, Serial T1 = 64). Administrative distance = **110**. It uses multicast 224.0.0.5 (all OSPF routers) and 224.0.0.6 (DR/BDR), IP protocol 89, supports VLSM, areas and fast convergence.

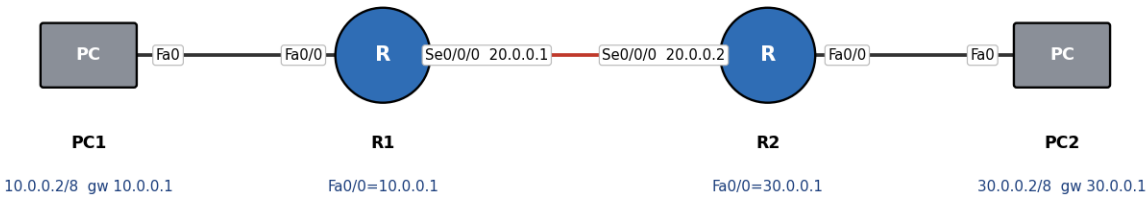
OSPF term	Meaning
Process ID	Locally significant number (1-65535) in *router ospf 1*; need not match on neighbours
Area	Logical group of routers; **Area 0** is the backbone and every other area must connect to it
Router ID	Unique 32-bit ID (highest loopback IP, else highest interface IP, or manually set with *router-id*)
Hello / Dead interval	10 s / 40 s on broadcast and point-to-point (30 / 120 on NBMA). **Must match** on neighbours
DR / BDR	Designated / Backup Designated Router elected on broadcast networks (Ethernet) to reduce LSA flooding; not used on point-to-point serial
Neighbour states	Down - Init - 2-Way - ExStart - Exchange - Loading - **Full**

Topology



OSPF topology from the manual

Exp 8: OSPF topology (red = serial DCE/DTE; Fa links = cross-over)



Topology with interfaces

Device	Interface	IP address	Mask	Gateway
R1 (2811)	Fa0/0	10.0.0.1	255.0.0.0	-

Device	Interface	IP address	Mask	Gateway
R1	Se0/0/0	20.0.0.1	255.0.0.0	-
R2 (2811)	Se0/0/0	20.0.0.2	255.0.0.0	-
R2	Fa0/0	30.0.0.1	255.0.0.0	-
PC1	Fa0	10.0.0.2	255.0.0.0	10.0.0.1
PC2	Fa0	30.0.0.2	255.0.0.0	30.0.0.1

NOTE: The 2811 router needs a serial card: switch the router **off**, drag **WIC-2T** into a free slot, switch it on (it gives Se0/0/0 and Se0/0/1). Connect with a **Serial DCE** cable. The DCE end (the end showing the clock icon in the manual figure is R2) needs *clock rate*; confirm with *show controllers serial 0/0/0*. PC to router links use **Copper Cross-Over**.

Step-by-step

Step 1: Build the network and configure PCs

Place 2 routers, 2 PCs, add WIC-2T cards, connect them. PC1: 10.0.0.2 / 255.0.0.0 / gw 10.0.0.1. PC2: 30.0.0.2 / 255.0.0.0 / gw 30.0.0.1.

Step 2: Configure R1 (interfaces + OSPF)

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface fastEthernet 0/0
R1(config-if)# ip address 10.0.0.1 255.0.0.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 20.0.0.1 255.0.0.0
R1(config-if)# no shutdown
R1(config-if)# exit
!  --- OSPF ---
R1(config)# router ospf 1                              ! process id 1
R1(config-router)# router-id 1.1.1.1                   ! optional but recommended
R1(config-router)# network 10.0.0.0 0.255.255.255 area 0
R1(config-router)# network 20.0.0.0 0.255.255.255 area 0
R1(config-router)# end
R1# copy running-config startup-config
```

Step 3: Configure R2 (interfaces + OSPF)

```
Router> enable
Router# configure terminal
Router(config)# hostname R2
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 20.0.0.2 255.0.0.0
R2(config-if)# clock rate 64000                        ! only on the DCE end
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface fastEthernet 0/0
R2(config-if)# ip address 30.0.0.1 255.0.0.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 20.0.0.0 0.255.255.255 area 0
R2(config-router)# network 30.0.0.0 0.255.255.255 area 0
R2(config-router)# end
R2# copy running-config startup-config
```

Explanation: the *network* command uses a **wildcard mask** (inverse of the subnet mask: 255.0.0.0 becomes 0.255.255.255). Any interface whose IP falls inside that range is enabled for OSPF in the given area and its network is advertised.

Testing / verification

Step 1: Neighbour adjacency (wait ~40 s)

```
R1# show ip ospf neighbor
Neighbor ID   Pri  State     Dead Time  Address     Interface
2.2.2.2         0  FULL/  -  00:00:34   20.0.0.2    Serial0/0/0
```

State must be **FULL**. Pri 0 and '-' (no DR) are normal on a point-to-point serial link.

Step 2: Routing table

```
R1# show ip route
C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, Serial0/0/0
O    30.0.0.0/8 [110/65] via 20.0.0.2, 00:01:10, Serial0/0/0

R1# show ip route ospf            ! only OSPF routes
R1# show ip protocols             ! OSPF process, router-id, networks, AD 110
R1# show ip ospf                  ! process details, area, SPF runs
R1# show ip ospf interface serial 0/0/0    ! cost, hello/dead timers, network type
R1# show ip ospf database         ! link-state database
```

O = OSPF route. Cost 65 = 64 (serial 1.544 Mbps) + 1 (FastEthernet on R2).

Step 3: End-to-end ping and traceroute

```
PC1> ping 30.0.0.2
PC1> tracert 30.0.0.2
  1   0 ms   0 ms   0 ms   10.0.0.1       (R1)
  2   1 ms   1 ms   1 ms   20.0.0.2       (R2)
  3   2 ms   2 ms   2 ms   30.0.0.2       (PC2)
PC2> ping 10.0.0.2
```

Troubleshooting OSPF (practise these deliberately)

Fault introduced	Symptom	Diagnose with	Fix
Wrong network statement (e.g. network 20.0.0.0 area 1 on one router)	No neighbour; area mismatch	*show ip protocols*, *debug ip ospf adj*	Use the same area on both ends
Hello/dead timers different (*ip ospf hello-interval 5* on one side)	Neighbour stuck / not formed	*show ip ospf interface*	*ip ospf hello-interval 10*, *ip ospf dead-interval 40* on both
Subnet mask mismatch on the link	Neighbour not formed	*show ip interface brief*	Same mask on both ends
Missing clock rate / interface shut	Serial down, no neighbour	*show ip interface brief*, *show controllers*	*clock rate 64000*; *no shutdown*
Duplicate router-id	Routes missing / flapping	*show ip ospf*	Give unique router-id and *clear ip ospf process*
LAN network not advertised	PC of that LAN unreachable	*show ip route* on the far router	Add the *network* statement for the LAN
Passive interface on the link	No neighbour on that link	*show ip protocols*	*no passive-interface*

```
R1# debug ip ospf events          ! see hellos and adjacency messages
R1# debug ip ospf adj
R1# undebug all
R1# clear ip ospf process         ! restart OSPF after a change (answer yes)
```

Conclusion

OSPF was configured on two routers, the neighbour state reached FULL, routes were exchanged and communication between the two LANs worked. Typical errors (wrong network statement, mismatched timers/areas/masks, down interfaces) were identified and corrected with the show and debug commands.

Viva Questions with Answers

Q1. What is the main difference between OSPF and RIP?

Ans: OSPF is link-state (full topology map, SPF/Dijkstra, cost metric, areas, fast convergence, no hop limit); RIP is distance-vector (hop count, periodic full-table updates, 15-hop limit, slow convergence).

Q2. How does OSPF calculate the best route?

Ans: Each link has a cost (reference bandwidth / interface bandwidth). Every router runs Dijkstra's SPF on its link-state database and selects the path with the lowest total cost.

Q3. What is the purpose of the OSPF area and why is Area 0 important?

Ans: Areas divide a large network to limit LSA flooding and SPF calculations. Area 0 is the backbone; all other areas must connect to it so that routes can be exchanged.

Q4. Explain the difference between OSPF broadcast and non-broadcast networks.

Ans: Broadcast (Ethernet): multicast hellos, DR/BDR election, hello 10 s/dead 40 s. Non-broadcast (NBMA, e.g. Frame Relay): no multicast so neighbours must be configured manually, DR/BDR election needed, hello 30 s/dead 120 s.

Q5. What is the hello and dead interval in OSPF, and why are they important?

Ans: Hello = how often hellos are sent (10 s); dead = time without hello after which the neighbour is declared down (40 s). They must match on both sides or no adjacency forms; they also control failure-detection time.

Practical 9 - VLAN Configuration and Troubleshooting

Aim

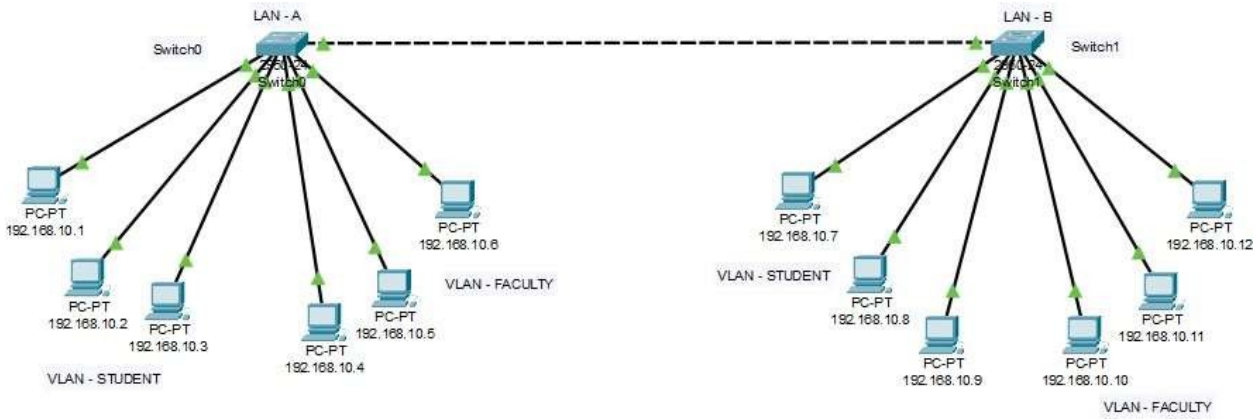
To configure VLANs on switches, assign ports to different VLANs, configure a trunk between switches, verify the segmentation and troubleshoot VLAN problems.

Theory

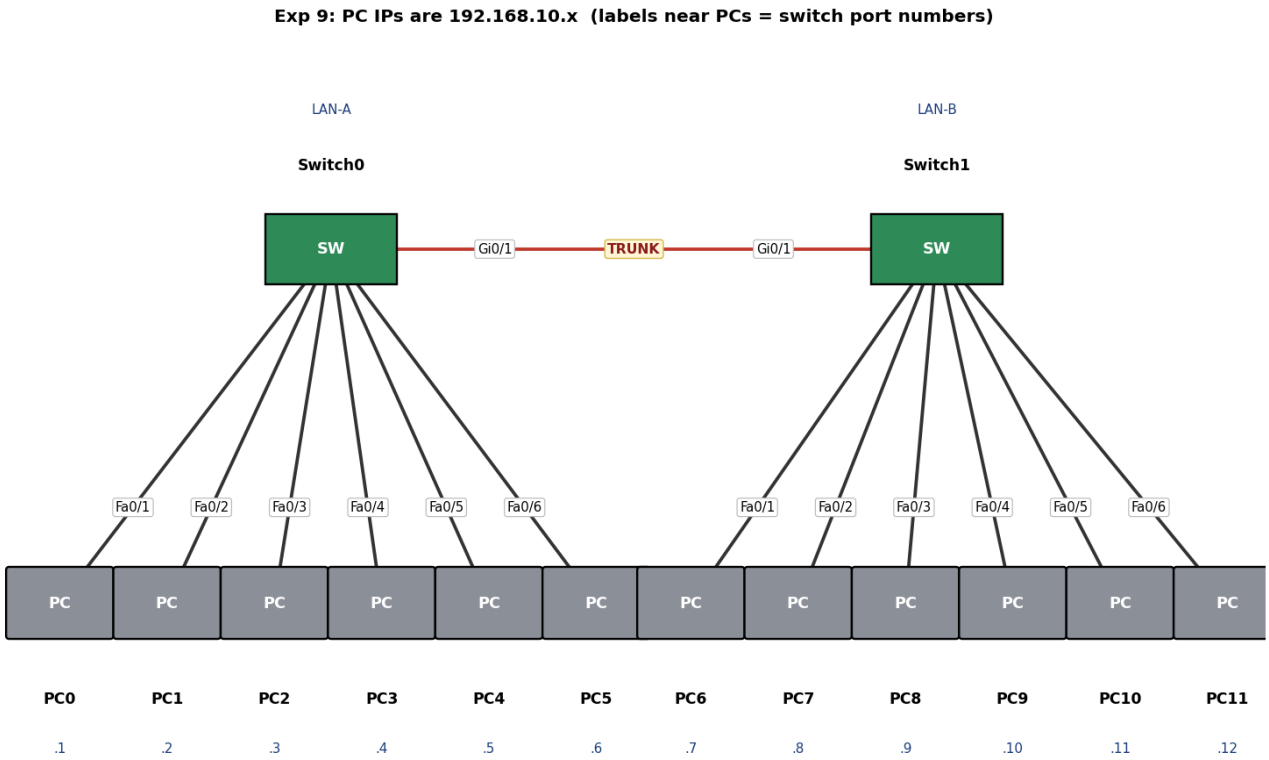
A **VLAN** is a logical grouping of devices that forms a separate **broadcast domain** on one or more switches, independent of physical location. It works at Layer 2. Devices of the same VLAN communicate directly; communication between different VLANs needs a Layer-3 device (router or Layer-3 switch). Advantages: security, smaller broadcast domains / better performance, easy management.

Term	Meaning
Access port	Belongs to exactly one VLAN; carries untagged frames; connects PCs
Trunk port	Carries frames of many VLANs between switches using IEEE 802.1Q tags (4-byte tag with VLAN ID)
Native VLAN	VLAN whose frames are sent untagged on a trunk (default VLAN 1); must match on both ends
Default VLAN	VLAN 1: all ports are in it initially
Data / Voice / Management VLAN	Different purposes: user traffic, prioritised IP-phone traffic, device management

Topology



VLAN topology from the manual (LAN-A and LAN-B, VLAN STUDENT and VLAN FACULTY)



Port assignment used in this guide

Device	Port	VLAN	IP address (mask 255.255.255.0)
Switch0: PC0, PC1, PC2	Fa0/1 - Fa0/3	10 STUDENT	192.168.10.1, .2, .3
Switch0: PC3, PC4, PC5	Fa0/4 - Fa0/6	20 FACULTY	192.168.10.4, .5, .6
Switch1: PC6, PC7, PC8	Fa0/1 - Fa0/3	10 STUDENT	192.168.10.7, .8, .9
Switch1: PC9, PC10, PC11	Fa0/4 - Fa0/6	20 FACULTY	192.168.10.10, .11, .12
Switch0 - Switch1	Gi0/1 - Gi0/1	Trunk	Copper Cross-Over cable

NOTE: The manual gives all 12 PCs addresses in the same network 192.168.10.0/24. That is fine for the VLAN demo: students can reach students, faculty can reach faculty, but a student cannot reach a faculty PC even though the IP network is the same - the switch itself blocks it because they are in different broadcast domains.

Step-by-step

Step 1: Build the topology

Place 2 switches (2960) and 12 PCs. Connect PCs to ports as in the table (Copper Straight-Through). Connect Switch0 Gi0/1 to Switch1 Gi0/1 with a **Copper Cross-Over**. Rename switches to Switch0 / Switch1 (optional).

Step 2: Assign IP addresses to the PCs

Use the table: PC0 = 192.168.10.1 ... PC11 = 192.168.10.12, mask 255.255.255.0 (no gateway needed).

Step 3: Configure Switch0

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch0
!  --- create the VLANs ---
Switch0(config)# vlan 10
Switch0(config-vlan)# name STUDENT
Switch0(config-vlan)# exit
Switch0(config)# vlan 20
Switch0(config-vlan)# name FACULTY
Switch0(config-vlan)# exit
!  --- assign access ports ---
Switch0(config)# interface range fastEthernet 0/1 - 3
```

```
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 10
Switch0(config-if-range)# exit
Switch0(config)# interface range fastEthernet 0/4 - 6
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 20
Switch0(config-if-range)# exit
!  --- trunk to the other switch ---
Switch0(config)# interface gigabitEthernet 0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# end
Switch0# copy running-config startup-config
```

Step 4: Configure Switch1 (identical, only the hostname differs)

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch1
Switch1(config)# vlan 10
Switch1(config-vlan)# name STUDENT
Switch1(config-vlan)# vlan 20
Switch1(config-vlan)# name FACULTY
Switch1(config-vlan)# exit
Switch1(config)# interface range fastEthernet 0/1 - 3
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 10
Switch1(config-if-range)# exit
Switch1(config)# interface range fastEthernet 0/4 - 6
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 20
Switch1(config-if-range)# exit
Switch1(config)# interface gigabitEthernet 0/1
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# end
Switch1# copy running-config startup-config
```

Important: the VLANs must exist on **both** switches (or use VTP). The 2960 supports only 802.1Q so there is no *switchport trunk encapsulation* command.

Testing / verification

Step 1: Show commands

```
Switch0# show vlan brief
VLAN Name              Status    Ports
---- ----------        --------- -------------------------------------
1    default           active    Fa0/7, Fa0/8, ... Fa0/24, Gig0/2
10   STUDENT           active    Fa0/1, Fa0/2, Fa0/3
20   FACULTY           active    Fa0/4, Fa0/5, Fa0/6
1002 fddi-default ... (default system VLANs)

Switch0# show interfaces trunk
Port     Mode   Encapsulation  Status    Native vlan
Gig0/1   on     802.1q         trunking  1
Port     Vlans allowed on trunk
Gig0/1   1-1005

Switch0# show interfaces fastEthernet 0/1 switchport
Switch0# show mac address-table
```

Step 2: Ping matrix (use PC Command Prompt)

From	To	Expected	Reason
PC0 (192.168.10.1)	PC1 (.2)	SUCCESS	Same VLAN 10, same switch
PC0 (.1)	PC6 (.7)	SUCCESS	Same VLAN 10 across the trunk
PC3 (.4)	PC9 (.10)	SUCCESS	Same VLAN 20 across the trunk
PC0 (.1)	PC3 (.4)	FAIL (timed out)	Different VLANs, no router
PC6 (.7)	PC9 (.10)	FAIL (timed out)	Different VLANs on the other switch

TIP: The first ping across the trunk may time out once because of ARP / MAC learning. Run it again.

Step 3: Simulation mode check

Send a simple PDU from PC0 to PC6 - it travels Switch0, trunk (tagged), Switch1 to PC6 only. A broadcast (ARP) from PC0 reaches only VLAN 10 ports (PC1, PC2, PC6, PC7, PC8) and never the faculty PCs. This proves broadcast-domain separation.

Troubleshooting VLAN problems

Symptom	Cause	Check	Fix
Same-VLAN PCs on different switches cannot ping	Trunk down / port not trunk	*show interfaces trunk*	*switchport mode trunk* on both ends, check cable type
One PC isolated from its VLAN	Port in wrong VLAN	*show vlan brief*	*switchport access vlan 10* on the port
VLAN missing on the far switch	VLAN not created there	*show vlan brief*	Create VLAN (same ID) on both switches
Port shows nothing / red light	Port shutdown or bad cable	*show interfaces status*	*no shutdown*, fix cable
Trunk up but one VLAN blocked	Allowed VLAN list / native VLAN mismatch	*show interfaces trunk*	*switchport trunk allowed vlan add 10,20*; same native VLAN on both ends

Extension: Inter-VLAN routing (router-on-a-stick) - answer to a viva question

To let STUDENT and FACULTY talk to each other, use different IP subnets (STUDENT 192.168.10.0/24, FACULTY 192.168.20.0/24; change the faculty PCs to 192.168.20.x with gateway 192.168.20.1 and students to gateway 192.168.10.1), add a router, connect its Gi0/0 to **Switch0 Fa0/24** and configure:

```
! --- Switch0 ---
Switch0(config)# interface fastEthernet 0/24
Switch0(config-if)# switchport mode trunk
! --- Router ---
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# end
```

Now **ping 192.168.20.x** from a student PC works (through the router). Check with *show ip route* (two connected networks).

Conclusion

VLAN 10 (STUDENT) and VLAN 20 (FACULTY) were created, ports were assigned and a trunk carried both VLANs between the switches. Devices in the same VLAN communicated, devices of different VLANs were isolated, and common VLAN faults were diagnosed.

Viva Questions with Answers

Q1. How does a switch handle VLAN traffic?

Ans: It keeps a separate forwarding domain for each VLAN: frames entering an access port are associated with that port's VLAN and are forwarded only to ports of the same VLAN (tagged with 802.1Q on trunks).

Q2. What is the difference between an access port and a trunk port?

Ans: Access port: one VLAN, untagged frames, connects end hosts. Trunk port: carries multiple VLANs with 802.1Q tags, connects switches/routers.

Q3. How do you configure inter-VLAN routing?

Ans: Use a router with sub-interfaces (router-on-a-stick: *encapsulation dot1Q <id>* + IP address per sub-interface) or a Layer-3 switch with SVIs (*interface vlan x* + *ip routing*).

Q4. What command would you use to verify VLANs configured on a switch?

Ans: *show vlan brief* (and *show interfaces trunk*, *show interfaces fa0/1 switchport*).

Q5. Why is trunking necessary when connecting multiple switches?

Ans: Without trunking each VLAN would need its own cable between the switches. One trunk carries all VLANs, tagged so the receiving switch knows the VLAN of each frame.

Q6. What is the difference between tagged and untagged VLAN traffic?

Ans: Tagged frames carry a 4-byte 802.1Q header with the VLAN ID (used on trunks); untagged frames have no tag (access ports and the native VLAN of a trunk).

Practical 10 - NAT Configuration and Troubleshooting

Aim

To implement Static NAT, Dynamic NAT and PAT (NAT overload) on a router, verify the translations and troubleshoot NAT.

Theory

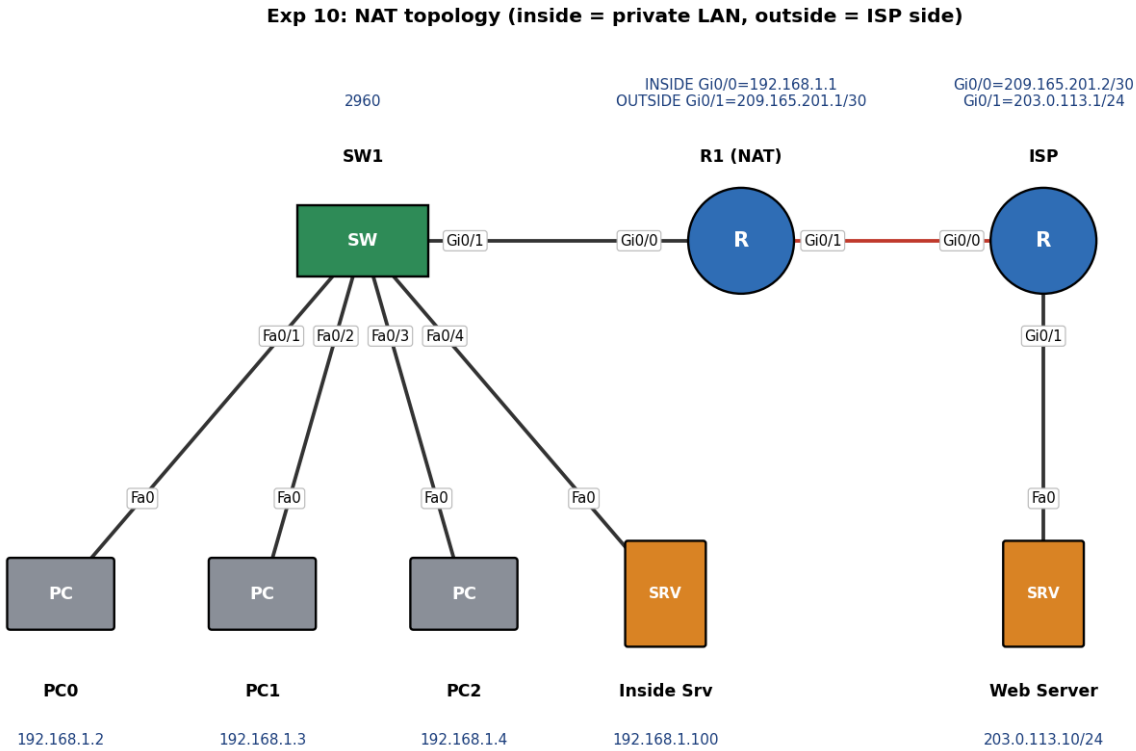
Network Address Translation converts private IPv4 addresses (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) to public addresses so many internal hosts can reach the Internet with few public addresses. It also hides the internal addressing.

Type	Mapping	Use
Static NAT	One private IP <-> one fixed public IP (1:1)	Internal server that must be reachable from outside
Dynamic NAT	Private IPs <-> pool of public IPs, first come first served	Limited public addresses; fails when the pool is exhausted
PAT / NAT overload	Many private IPs → one public IP, distinguished by port number	Home/office Internet access (most common)

Terms: *inside local* = private address of the inside host; *inside global* = public address representing it; *outside global* = address of the remote host. *ip nat inside* marks the LAN-facing interface, *ip nat outside* the Internet-facing interface. The **translation table** stores the mappings.

Topology (designed for this lab)

The manual gives no figure for NAT, so a standard topology is used: a private LAN behind the NAT router R1, an 'ISP' router, and a public web server.



NAT topology

Device	Interface	IP address / mask	Gateway
PC0	Fa0	192.168.1.2 / 255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.3 / 255.255.255.0	192.168.1.1

Device	Interface	IP address / mask	Gateway
PC2	Fa0	192.168.1.4 / 255.255.255.0	192.168.1.1
Inside Server	Fa0	192.168.1.100 / 255.255.255.0	192.168.1.1
R1 (2911)	Gi0/0 (inside)	192.168.1.1 / 255.255.255.0	-
R1	Gi0/1 (outside)	209.165.201.1 / 255.255.255.252	-
ISP (2911)	Gi0/0	209.165.201.2 / 255.255.255.252	-
ISP	Gi0/1	203.0.113.1 / 255.255.255.0	-
Web Server	Fa0	203.0.113.10 / 255.255.255.0	203.0.113.1

Public addresses used for NAT: dynamic pool **209.165.202.129 - 209.165.202.130** (only 2 addresses on purpose); static NAT address for the Inside Server **209.165.202.131**. Cabling: PCs/Inside Server to SW1 and SW1 Gi0/1 to R1 Gi0/0 (straight-through); R1 Gi0/1 to ISP Gi0/0 (**cross-over**); ISP Gi0/1 to Web Server (**cross-over**).

Step-by-step

Step 1: Build the topology and configure end devices

Set the IP address, mask and gateway of PC0, PC1, PC2, Inside Server and Web Server from the table. In the Web Server enable **Services > HTTP** (on by default) so a web page can be opened.

Step 2: Configure the ISP router (no NAT here)

```
Router> enable
Router# configure terminal
Router(config)# hostname ISP
ISP(config)# interface gigabitEthernet 0/0
ISP(config-if)# ip address 209.165.201.2 255.255.255.252
ISP(config-if)# no shutdown
ISP(config-if)# interface gigabitEthernet 0/1
ISP(config-if)# ip address 203.0.113.1 255.255.255.0
ISP(config-if)# no shutdown
ISP(config-if)# exit
!  route back to the public NAT addresses (needed for static/dynamic NAT)
ISP(config)# ip route 209.165.202.128 255.255.255.248 209.165.201.1
ISP(config)# end
ISP# copy running-config startup-config
```

Step 3: Configure R1 interfaces, default route and the ACL

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# ip nat inside                        ! LAN side
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 209.165.201.1 255.255.255.252
R1(config-if)# ip nat outside                       ! Internet side
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.2    ! all unknown traffic to ISP
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255   ! who may be translated
```

Part A - Static NAT

```
R1(config)# ip nat inside source static 192.168.1.100 209.165.202.131
R1(config)# end
```

Test: Inside Server > **ping 203.0.113.10** works. From the **Web Server** Desktop > Command Prompt: **ping 209.165.202.131** - it reaches the inside server (the outside host only knows the public address). On R1:

```
R1# show ip nat translations
Pro  Inside global      Inside local     Outside local    Outside global
---  209.165.202.131    192.168.1.100    ---              ---
```

Part B - Dynamic NAT (pool)

```
R1# configure terminal
R1(config)# no ip nat inside source static 192.168.1.100 209.165.202.131
R1(config)# ip nat pool PUBLIC-POOL 209.165.202.129 209.165.202.130 netmask 255.255.255.248
R1(config)# ip nat inside source list 1 pool PUBLIC-POOL
R1(config)# end
R1# clear ip nat translation *
```

Test: PC0 > **ping 203.0.113.10**, PC1 > **ping 203.0.113.10** - both succeed. Now ping from **PC2**: it **fails** because only 2 public addresses exist (pool exhausted) while the other two translations are still active. Check:

```
R1# show ip nat translations
Pro  Inside global      Inside local   Outside local   Outside global
icmp 209.165.202.129:1  192.168.1.2:1  203.0.113.10:1  203.0.113.10:1
icmp 209.165.202.130:1  192.168.1.3:1  203.0.113.10:1  203.0.113.10:1
R1# show ip nat statistics         ! shows misses, pool usage
```

Part C - PAT / NAT overload (most common)

```
R1# configure terminal
R1(config)# no ip nat inside source list 1 pool PUBLIC-POOL
R1(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
R1(config)# end
R1# clear ip nat translation *
```

The ISP does not need the extra route for PAT because the translated source is the outside interface 209.165.201.1, which is directly connected to the ISP. **Test:** ping or open the web page **http://203.0.113.10** (PC > Desktop > Web Browser) from **all three PCs** - all work now.

```
R1# show ip nat translations
Pro  Inside global         Inside local      Outside local     Outside global
icmp 209.165.201.1:1       192.168.1.2:1     203.0.113.10:1    203.0.113.10:1
icmp 209.165.201.1:2       192.168.1.3:2     203.0.113.10:2    203.0.113.10:2
tcp  209.165.201.1:1025    192.168.1.4:1025  203.0.113.10:80   203.0.113.10:80

R1# show ip nat statistics
Total translations: 3 (0 static, 3 dynamic, 3 extended)
Outside interfaces: GigabitEthernet0/1    Inside interfaces: GigabitEthernet0/0
Hits: 25  Misses: 0
R1# debug ip nat                  ! live translation messages (undebug all to stop)
```

All inside hosts share the single address **209.165.201.1**; different **port numbers** keep the sessions apart. Use **Simulation mode** and open the PDU at R1: the 'Inside/Outside' source address changes from 192.168.1.x to 209.165.201.1 (PDU information > Outbound PDU Details).

NAT troubleshooting checklist

Symptom	Possible cause	Fix
No translations in table	*ip nat inside/outside* missing or swapped on interfaces	Re-apply on the correct interfaces (*show ip nat statistics* lists them)
Ping from LAN fails, no translation	ACL does not match the inside network	*show access-lists*; ACL 1 must permit 192.168.1.0 0.0.0.255
Translation exists, reply never returns	ISP lacks route to the public pool	Add *ip route 209.165.202.128 255.255.255.248 209.165.201.1* on ISP
Some hosts work, others fail (dynamic NAT)	Pool exhausted	Larger pool or use *overload* (PAT)
Stale / wrong mappings after changes	Old entries remain	*clear ip nat translation **
Nothing leaves R1	No default route	*ip route 0.0.0.0 0.0.0.0 209.165.201.2*

Conclusion

Static NAT, dynamic NAT and PAT were implemented. Private addresses were translated to public addresses, the translation table was examined with *show ip nat translations* and common NAT problems were troubleshot. NAT conserves public IPv4 addresses and hides the internal network.

Viva Questions with Answers

Q1. What is the primary purpose of NAT in networking?

Ans: To translate private IP addresses to public ones so that many hosts can share few public addresses (conserving IPv4 space) and to hide the internal addressing.

Q2. Explain the difference between static NAT and dynamic NAT.

Ans: Static NAT: permanent one-to-one mapping (used for servers). Dynamic NAT: temporary mapping from a pool of public addresses to inside hosts on demand.

Q3. How does PAT work, and why is it useful?

Ans: PAT (overload) maps many private addresses to one public address, using unique source port numbers to tell sessions apart. Useful because one public IP serves the whole LAN.

Q4. What are the advantages and disadvantages of using NAT?

Ans: Advantages: saves public IPs, hides internal network, flexible addressing. Disadvantages: breaks end-to-end connectivity, adds delay/processing, complicates some protocols (IPsec, VoIP) and logging/troubleshooting.

Q5. Describe the steps involved in configuring NAT on a router.

Ans: (1) Mark inside and outside interfaces; (2) create an ACL for inside addresses; (3) define a pool (dynamic) or a static mapping or choose the interface for overload; (4) bind ACL to pool/interface with *ip nat inside source*; (5) verify with show ip nat translations/statistics.

Practical 11 - Network Utility Tools: Nmap, Wireshark, Network Scanner

IMPORTANT: Use these tools **only on your own lab network or on systems you have written permission to test**. Scanning others without permission may be illegal. For safe practice Nmap provides the host **scanme.nmap.org** for testing.

11.1 Nmap

Aim

To discover hosts and services on a network and assess security by finding open ports and running services.

Theory

Nmap (Network Mapper) is an open-source tool for network exploration and security auditing. It sends crafted packets and analyses the replies to perform host discovery, port scanning, service/version detection and OS detection. Port states: **open** (a service accepts connections), **closed** (reachable but no service), **filtered** (firewall drops probes, state unknown).

Step-by-step

Step 1: Install

Windows: download the installer from nmap.org (it also installs Zenmap GUI). Linux: **sudo apt install nmap**.

Step 2: Find your network

```
C:\> ipconfig          (Linux: ip addr)       -> e.g. 192.168.1.10 / 255.255.255.0 => network 192.168.1.0/24
```

Step 3: Run the scans and note the results

Command	What it does
nmap -sn 192.168.1.0/24	Host discovery (ping scan): lists live hosts, no port scan
nmap 192.168.1.1	Scan the 1000 most common TCP ports of one host
nmap -p 22,80,443 192.168.1.1	Scan only the given ports
nmap -p 1-65535 192.168.1.1	Scan all ports
nmap -sT 192.168.1.1	TCP connect scan (no admin rights needed)
nmap -sS 192.168.1.1	SYN (half-open) scan (needs admin/root)
nmap -sU -p 53,67,161 192.168.1.1	UDP scan of selected ports
nmap -sV 192.168.1.1	Detect service names and versions
nmap -O 192.168.1.1	Operating-system detection (admin/root)
nmap -A scanme.nmap.org	Aggressive: OS, version, scripts, traceroute
nmap -Pn 192.168.1.1	Skip host discovery (host blocks ping)
nmap -oN result.txt 192.168.1.0/24	Save output to a file (-oX for XML)

```
$ nmap -sV scanme.nmap.org
PORT      STATE  SERVICE  VERSION
22/tcp    open   ssh      OpenSSH 6.6.1
80/tcp    open   http     Apache httpd 2.4.7
9929/tcp  open   nping-echo
```

Algorithm: input target(s) → send TCP/UDP/ICMP/ARP probes → analyse responses to classify ports → optional service and OS probing → generate report. **Output:** list of live devices, open ports and states, service versions, OS details.

Conclusion

Nmap gives a clear picture of the hosts, open ports and services of a network and is used by administrators and security professionals to detect unnecessary open ports and weaknesses.

11.2 Wireshark

Aim

To capture and analyse network traffic in real time for troubleshooting, protocol analysis and security monitoring.

Theory

Wireshark is a protocol analyser. It puts the network card in promiscuous mode, captures frames, and decodes them layer by layer (Ethernet, IP, TCP/UDP, HTTP, DNS...). **Capture filters** (BPF syntax, set before capture, e.g. *host 192.168.1.1*) limit what is stored; **display filters** (e.g. *ip.addr == 192.168.1.1*) hide packets after capture.

Step-by-step

Step 1: Select the interface and start

Open Wireshark, double-click your active interface (Wi-Fi or Ethernet - the one with the moving graph). Capture starts at once.

Step 2: Generate traffic and apply display filters

Display filter	Shows
icmp	Ping packets (run *ping 8.8.8.8* in cmd and watch Echo request / reply)
arp	ARP requests/replies (who has 192.168.1.1?)
dns	DNS queries and responses (run *nslookup google.com*)
http	Unencrypted web traffic
tcp.port == 80	Traffic to/from port 80
ip.addr == 192.168.1.1	Packets to or from that host
tcp.flags.syn == 1 && tcp.flags.ack == 0	TCP connection starts (first packet of the 3-way handshake)
!(arp || dns)	Everything except ARP and DNS

Step 3: Analyse a packet

Click a packet: the middle pane shows the layers (Frame, Ethernet II - MAC addresses, IPv4 - source/destination IP and TTL, TCP/UDP - ports, flags, sequence numbers, application data). The bottom pane shows raw hex. Right-click a TCP packet > **Follow > TCP Stream** to read the whole conversation.

Step 4: Statistics and saving

Statistics > Protocol Hierarchy (protocol distribution), **Conversations**, **I/O Graph**. Stop with the red square and **File > Save As** (.pcapng). **File > Export Packet Dissections** exports to CSV/text.

Mini exercises

- Capture a **TCP three-way handshake**: open a website, filter *tcp.flags.syn==1*, see SYN, SYN-ACK, ACK.
- Capture the **DHCP DORA** exchange: filter *bootp* (or *dhcp*), run *ipconfig /release* and */renew*.
- Show why HTTP is insecure: log in on a plain-HTTP test page and find the password in the packet bytes.

Conclusion

Wireshark allows detailed packet-level analysis, useful for troubleshooting, performance monitoring and finding security issues.

11.3 Network Scanner (general)

Aim

To discover all devices of a network with their IP and MAC addresses and find the services running on them.

Theory

A network scanner sends ARP requests (same LAN) or ICMP echo requests (routed) to every address of a range and lists the hosts that answer; optionally it port-scans them and resolves the manufacturer from the MAC (OUI) prefix. Examples: Advanced IP Scanner, Angry IP Scanner, Nmap, arp-scan.

Step-by-step (Advanced IP Scanner / Angry IP Scanner)

Step 1: Install and start the scanner.

Step 2: Enter the IP range, e.g. 192.168.1.1 - 192.168.1.254 (the program often fills it automatically).

Step 3: Click Scan.

Step 4: Read the table: Status, IP, Name, Manufacturer (from MAC), MAC address, open services (HTTP, FTP, shared folders).

Step 5: Compare with the list of devices you expect; unknown devices need investigation (unauthorised devices).

Step 6: Save/export the result (CSV/HTML).

Command-line alternative:

```
C:\> arp -a                       ! devices that your PC already talked to
$ sudo arp-scan --localnet         ! Linux, ARP-based scan of the local network
$ nmap -sn 192.168.1.0/24          ! discovery scan, prints IP + MAC + vendor
```

Conclusion

Network scanners map the network, reveal unauthorised devices and open services, helping in network management and security auditing.

Viva Questions with Answers

Q1. What is the primary purpose of Nmap, and how does it work?

Ans: Nmap discovers hosts and services (open ports, versions, OS). It sends probe packets (ICMP, ARP, TCP SYN, UDP) and classifies ports from the responses.

Q2. How does Wireshark differ from Nmap in functionality?

Ans: Wireshark passively captures and decodes traffic that passes the interface (analysis); Nmap actively sends probes to discover hosts/ports (scanning).

Q3. Describe the steps involved in using a network scanner.

Ans: Enter IP range, start the scan (ARP/ICMP), collect responses, optionally port-scan, view/save the list of devices.

Q4. What types of information can Nmap provide during a scan?

Ans: Live hosts, open/closed/filtered ports, service names and versions, OS guess, MAC vendor, script results (vulnerabilities), route.

Q5. How can you filter packets in Wireshark, and why is this useful?

Ans: Capture filters (before capture) and display filters (after capture, e.g. tcp.port==80). Filtering removes noise so you can focus on the relevant traffic.

Q6. Explain the significance of open ports in network security assessments.

Ans: Each open port is a possible entry point; unnecessary or outdated services increase the attack surface, so open ports are reviewed, closed or protected.

Q7. What are some ethical considerations when using these tools?

Ans: Obtain written authorisation, limit the scope, avoid disruption, protect captured data and privacy, and report findings responsibly.

Q8. Discuss a scenario where Wireshark would be more beneficial than Nmap.

Ans: Finding why an application is slow or why DHCP/DNS fails - you need to see the actual packets (retransmissions, missing replies), not just port states.

Q9. How can network utilities assist in penetration testing?

Ans: They support reconnaissance (Nmap, scanners), traffic analysis (Wireshark), and verification of fixes in the authorised test.

Q10. What are the limitations of these tools in a corporate environment?

Ans: Firewalls/IDS may block or detect scans, switched networks limit what Wireshark sees (needs SPAN/mirror port), encryption hides payloads, and policy/legal approval is required.

Practical 12 - Security: Threats, Attacks, Mitigation and Device Security

12.1 Security Threats and Vulnerabilities

Aim

To identify the security threats and vulnerabilities that affect networks and devices and to plan how to reduce the risk.

Theory

A **vulnerability** is a weakness (unpatched software, default password, open port); a **threat** is something that can exploit it (malware, hacker, insider); **risk** = likelihood x impact. Security aims at the CIA triad: **Confidentiality, Integrity, Availability**.

Threat	Description	Typical vulnerability exploited
Malware (virus, worm, ransomware, trojan)	Malicious software that damages or steals data	Missing antivirus / patches, unsafe downloads
Phishing	Fake mails/sites that trick users into giving credentials	Untrained users
DDoS	Flood of traffic makes a service unavailable	No rate limiting / filtering
Insider threat	Employee misuses access	Excess privileges, no logging
Weak / default passwords	Easy to guess or known credentials	Devices left with default passwords
Unpatched software	Known bugs exploited	No patch management

Algorithm (risk assessment)

- Identify assets (data, devices, networks).
- Assess vulnerabilities (vulnerability scan, e.g. Nmap scripts, Nessus).
- Analyse threats that could exploit them.
- Evaluate the impact.
- Prioritise risks by severity x likelihood and plan controls.

Output: list of vulnerabilities and threats, risk-assessment report and a prioritised list of security measures.

12.2 Network Attacks

Aim

To categorise and describe network attacks that threaten confidentiality, integrity and availability.

Attack	How it works	Impact	Example / defence
Denial of Service (DoS/DDoS)	Overwhelms a server/link with traffic or half-open connections (SYN flood)	Availability lost	Rate limiting, firewall, CDN / scrubbing
Man-in-the-Middle (MitM)	Attacker sits between two parties (ARP spoofing, rogue Wi-Fi) and reads/alters traffic	Confidentiality, integrity	HTTPS/TLS, VPN, dynamic ARP inspection
Phishing	Deceptive e-mail/site collects credentials	Credential theft	User training, MFA, mail filtering
Malware / Ransomware	Executes malicious code, encrypts files	Data loss	Antivirus, backups, patching
Password attacks	Brute force / dictionary	Account takeover	Strong passwords, lock-out, MFA
Sniffing / spoofing	Capturing traffic; faking IP/MAC addresses	Information leak	Encryption, port security
DNS poisoning	False DNS records are cached	Users redirected to fake sites	DNSSEC, trusted resolvers

Output: a report with the attack types, mechanism, examples and the risk of each.

12.3 Network Attack Mitigation

Mitigation = security controls that reduce the likelihood or the impact of an attack: **firewalls**, **IDS/IPS**, access control lists, VPN, network segmentation (VLANs), port security, patching, monitoring/logging, backups and an **incident-response plan**. Steps: identify vulnerabilities → implement controls → prepare the incident response plan → monitor continuously → train staff.

12.4 Device Security

Steps: inventory all devices → assess security posture → apply controls → monitor logs → educate users.

Control	Examples
Endpoint protection	Antivirus / anti-malware, host firewall
Updates	Regular OS and firmware patches
Encryption	Disk encryption, HTTPS, SSH instead of Telnet
Secure configuration	Change default passwords, disable unused services/ports, strong passwords, MFA
Monitoring	Review device logs, syslog, alerts

Hands-on demo in Packet Tracer: securing a switch and router (mitigation)

Use the star topology of Practical 3 (SW1 with PCs on Fa0/1 - Fa0/6; switch management IP 192.168.1.100/24; PC0 = 192.168.1.1). These configurations implement the 'device security' and 'attack mitigation' controls.

Step 1: Replace Telnet with SSH (encrypted remote access) and use local users

```
SW1> enable
SW1# configure terminal
SW1(config)# ip domain-name lab.local
SW1(config)# username admin secret Admin@123
SW1(config)# crypto key generate rsa            ! enter 1024 when asked for modulus size
SW1(config)# ip ssh version 2
SW1(config)# line vty 0 15
SW1(config-line)# transport input ssh           ! block Telnet
SW1(config-line)# login local
SW1(config-line)# exec-timeout 5 0              ! auto-logout after 5 minutes
SW1(config-line)# exit
```

Test: PC0 > Command Prompt > **ssh -l admin 192.168.1.100** (works, asks for the password); **telnet 192.168.1.100** now fails.

Step 2: Allow management only from the admin PC (ACL on VTY)

```
SW1(config)# access-list 10 permit host 192.168.1.1
SW1(config)# line vty 0 15
SW1(config-line)# access-class 10 in
SW1(config-line)# exit
```

Test: SSH from PC0 works; SSH from PC1 (192.168.1.2) is refused.

Step 3: Port security against MAC flooding / unauthorised devices

```
SW1(config)# interface range fastEthernet 0/1 - 6
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport port-security
SW1(config-if-range)# switchport port-security maximum 1
SW1(config-if-range)# switchport port-security mac-address sticky      ! learn the first MAC
SW1(config-if-range)# switchport port-security violation shutdown
SW1(config-if-range)# exit
```

Test: ping from every PC once so the MACs are learned; then **show port-security interface fa0/1** and **show port-security address**. Disconnect PC0, connect a different PC/laptop to Fa0/1 and ping: the port goes **err-disabled** (red light) - the intruder is blocked. Recover with *interface fa0/1* > *shutdown* > *no shutdown*.

Step 4: Disable unused ports and add a warning banner

```
SW1(config)# interface range fastEthernet 0/7 - 24
SW1(config-if-range)# shutdown
SW1(config-if-range)# exit
SW1(config)# banner motd #Unauthorized access is prohibited#
SW1(config)# service password-encryption
SW1(config)# end
SW1# show running-config
SW1# copy running-config startup-config
```

Step 5: Packet filtering with an ACL on a router (firewall behaviour)

If the PCs are behind a router (e.g. Practical 6/10), block one host from reaching a server and allow everything else:

```
R1(config)# ip access-list extended BLOCK-PC
R1(config-ext-nacl)# deny ip host 192.168.1.3 host 203.0.113.10
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)# exit
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip access-group BLOCK-PC in
R1(config-if)# end
R1# show access-lists          ! match counters increase when packets are denied
```

IMPORTANT: Never use real attack tools outside an authorised lab. Packet Tracer demos here are defensive only.

Conclusion

Threats, vulnerabilities and attacks were studied and mitigation methods (SSH, ACL, port security, disabled ports, patching, monitoring, training, incident response) were explained and practised. Layered security protects the network and every device.

Viva Questions with Answers

Q1. What are the main types of security threats faced by networks?

Ans: Malware, phishing, DoS/DDoS, man-in-the-middle, insider threats, password attacks and exploitation of unpatched software.

Q2. How do vulnerabilities differ from threats in network security?

Ans: A vulnerability is a weakness in a system; a threat is an actor/event that can exploit it. Risk appears when a threat meets a vulnerability.

Q3. Explain the concept of a Denial of Service (DoS) attack and its impact.

Ans: An attacker floods a target with traffic or requests so legitimate users cannot be served, causing downtime and financial loss (attack on availability).

Q4. What are some effective strategies for mitigating network attacks?

Ans: Firewalls, IDS/IPS, ACLs, segmentation, patching, strong authentication/MFA, encryption (VPN, TLS, SSH), port security, monitoring and backups.

Q5. How can organizations protect their devices from security threats?

Ans: Antivirus, updates, encryption, secure configuration (no default passwords), access control, monitoring logs and user training.

Q6. What role does user education play in enhancing security?

Ans: Most attacks (phishing, weak passwords) target users; trained users recognise and report threats, so human errors drop.

Q7. Describe the importance of an incident response plan in network security.

Ans: It defines roles and steps to detect, contain, eradicate and recover from incidents quickly, reducing damage and downtime, and lets the organisation learn from incidents.

Q8. What tools can be used to detect and prevent network attacks?

Ans: Firewalls, IDS/IPS (Snort), SIEM, antivirus/EDR, vulnerability scanners (Nessus, Nmap), Wireshark for analysis.

Practical 13 - Case Study / Mini Project: Build a Small Network and Scale It

Aim

To design a small office network, scale it to a larger network with departments (VLANs), routing (OSPF), a file server and a backup path, and to troubleshoot typical problems.

Scenario (from the manual)

A start-up first has one router, one switch and two PCs with DHCP and file sharing. As it grows it needs two more departments (each with its own subnet), a file server, a redundant path, more devices and troubleshooting procedures. The case study has three phases:

Phase	What is done	Technologies
1. Small network	Router + switch + 2 PCs, DHCP, ping test	DHCP, ping
2. Scaling	3 departments, central file server, backup path, more users	VLANs, router-on-a-stick, OSPF, redundancy, ACL
3. Troubleshooting	Wrong VLAN, IP conflict, slow link	ping, tracert, show commands, DHCP exclusions, QoS

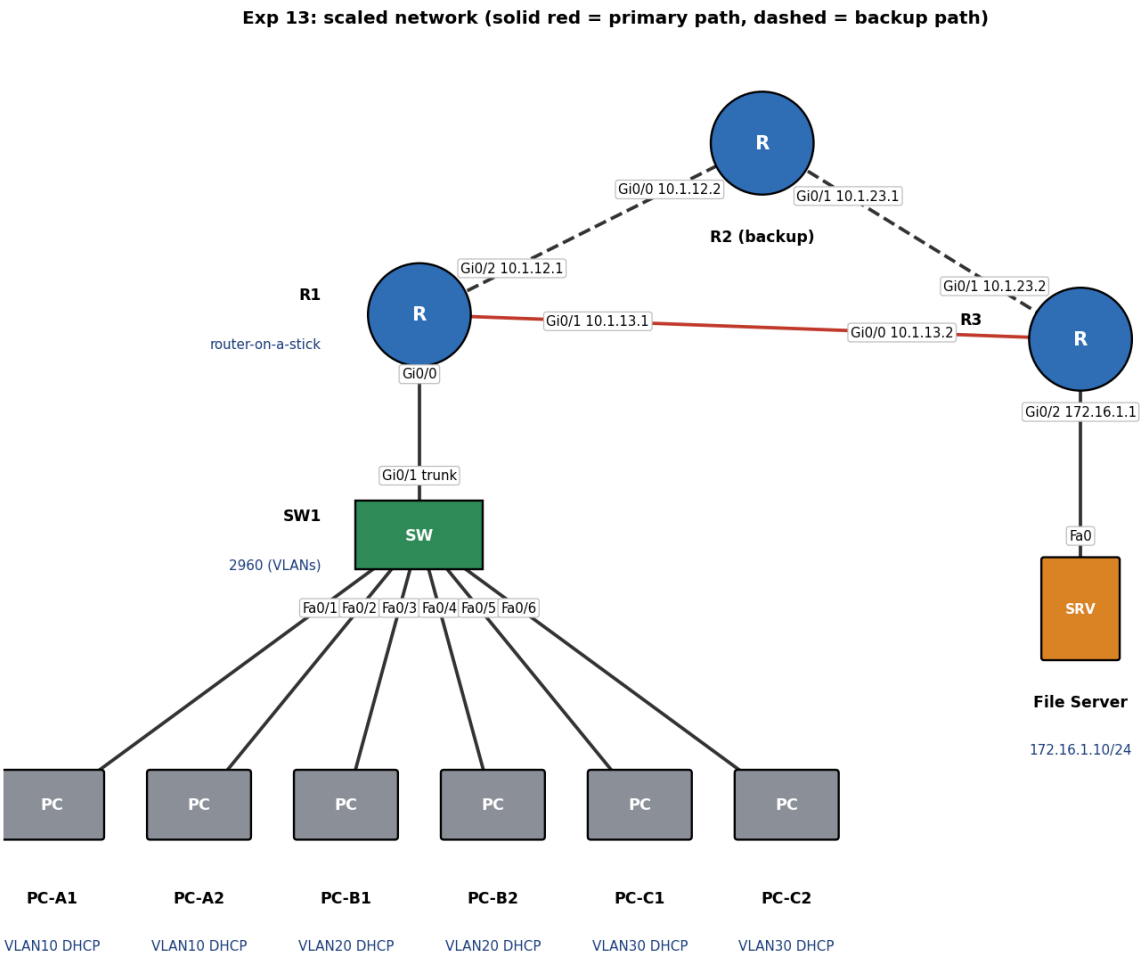
PHASE 1 - Small network

Place Router R1 (2911), Switch SW1 (2960), PC-A1, PC-A2. Connect R1 Gi0/0 to SW1 Gi0/1; PCs to Fa0/1, Fa0/2.

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)# ip dhcp pool DEPT-A
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# end
R1# copy running-config startup-config
```

Set both PCs to DHCP, then **ping 192.168.10.1** and ping each other. File sharing between PCs is demonstrated in real labs with SMB; in Packet Tracer it is represented by pinging and by the file server's FTP service in Phase 2.

PHASE 2 - Scaling the network



Scaled network: 3 VLAN departments, OSPF with primary and backup path to the file server

Item	Detail
VLAN 10 DEPT-A	192.168.10.0/24, gateway 192.168.10.1 (PC-A1, PC-A2 on Fa0/1-2)
VLAN 20 DEPT-B	192.168.20.0/24, gateway 192.168.20.1 (PC-B1, PC-B2 on Fa0/3-4)
VLAN 30 DEPT-C	192.168.30.0/24, gateway 192.168.30.1 (PC-C1, PC-C2 on Fa0/5-6)
R1 - R3 (primary)	10.1.13.0/30: R1 Gi0/1 = .1, R3 Gi0/0 = .2
R1 - R2 (backup)	10.1.12.0/30: R1 Gi0/2 = .1, R2 Gi0/0 = .2
R2 - R3 (backup)	10.1.23.0/30: R2 Gi0/1 = .1, R3 Gi0/1 = .2
File server LAN	172.16.1.0/24: R3 Gi0/2 = 172.16.1.1, File Server = 172.16.1.10, gateway 172.16.1.1

Cables: PCs to SW1 and SW1 Gi0/1 to R1 Gi0/0 = Copper Straight-Through; router-to-router and R3-to-server = Copper Cross-Over.

Step 1: Configure SW1 (VLANs, access ports, trunk to the router)

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
SW1(config)# vlan 10
SW1(config-vlan)# name DEPT-A
SW1(config-vlan)# vlan 20
SW1(config-vlan)# name DEPT-B
SW1(config-vlan)# vlan 30
SW1(config-vlan)# name DEPT-C
SW1(config-vlan)# exit
SW1(config)# interface range fastEthernet 0/1 - 2
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config-if-range)# interface range fastEthernet 0/3 - 4
SW1(config-if-range)# switchport mode access
```

```
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# interface range fastEthernet 0/5 - 6
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 30
SW1(config-if-range)# exit
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# end
SW1# copy running-config startup-config
```

Step 2: Configure R1 (sub-interfaces, DHCP, OSPF)

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no ip address                        ! remove the Phase-1 address
R1(config-if)# no shutdown
R1(config-if)# interface gigabitEthernet 0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# interface gigabitEthernet 0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# interface gigabitEthernet 0/0.30
R1(config-subif)# encapsulation dot1Q 30
R1(config-subif)# ip address 192.168.30.1 255.255.255.0
R1(config-subif)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 10.1.13.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# interface gigabitEthernet 0/2
R1(config-if)# ip address 10.1.12.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
! --- DHCP for the three departments ---
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.10
R1(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.10
R1(config)# ip dhcp pool DEPT-A
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# ip dhcp pool DEPT-B
R1(dhcp-config)# network 192.168.20.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.20.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# ip dhcp pool DEPT-C
R1(dhcp-config)# network 192.168.30.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.30.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# exit
! --- OSPF ---
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# network 192.168.20.0 0.0.0.255 area 0
R1(config-router)# network 192.168.30.0 0.0.0.255 area 0
R1(config-router)# network 10.1.13.0 0.0.0.3 area 0
R1(config-router)# network 10.1.12.0 0.0.0.3 area 0
R1(config-router)# end
R1# copy running-config startup-config
```

Step 3: Configure R2 (backup router)

```
Router> enable
Router# configure terminal
Router(config)# hostname R2
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 10.1.12.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# interface gigabitEthernet 0/1
R2(config-if)# ip address 10.1.23.1 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# router ospf 1
```

```
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.1.12.0 0.0.0.3 area 0
R2(config-router)# network 10.1.23.0 0.0.0.3 area 0
R2(config-router)# end
R2# copy running-config startup-config
```

Step 4: Configure R3 (server-side router) with file-server protection

```
Router> enable
Router# configure terminal
Router(config)# hostname R3
R3(config)# interface gigabitEthernet 0/0
R3(config-if)# ip address 10.1.13.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface gigabitEthernet 0/1
R3(config-if)# ip address 10.1.23.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface gigabitEthernet 0/2
R3(config-if)# ip address 172.16.1.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 10.1.13.0 0.0.0.3 area 0
R3(config-router)# network 10.1.23.0 0.0.0.3 area 0
R3(config-router)# network 172.16.1.0 0.0.0.255 area 0
R3(config-router)# exit
! --- secure access: DEPT-C must NOT reach the file server, all others may ---
R3(config)# ip access-list extended PROTECT-SERVER
R3(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 host 172.16.1.10
R3(config-ext-nacl)# permit ip any any
R3(config-ext-nacl)# exit
R3(config)# interface gigabitEthernet 0/2
R3(config-if)# ip access-group PROTECT-SERVER out
R3(config-if)# end
R3# copy running-config startup-config
```

File Server: IP 172.16.1.10, mask 255.255.255.0, gateway 172.16.1.1; enable **Services > FTP** (user cisco / password cisco).

Step 5: Set all six PCs to DHCP

Each PC: Desktop > IP Configuration > DHCP. PC-A1 should get 192.168.10.11, PC-B1 192.168.20.11, PC-C1 192.168.30.11, etc.

Testing the scaled network

Test	Command	Expected result
DHCP per VLAN	PC-B1: *ipconfig*	192.168.20.x, gateway 192.168.20.1
Inside VLAN	PC-A1 ping PC-A2	Success
Inter-VLAN routing	PC-A1 ping 192.168.20.11	Success (via R1 sub-interfaces)
Reach the file server	PC-A1 ping 172.16.1.10	Success; OSPF learned the route
FTP to the server	PC-A1 Command Prompt: *ftp 172.16.1.10*	Login succeeds
Security ACL	PC-C1 ping 172.16.1.10	**Fails** (denied by ACL on R3), but C1 can still ping A1
Primary path	PC-A1 *tracert 172.16.1.10*	192.168.10.1 - 10.1.13.2 - 172.16.1.10

```
R1# show ip route ospf
O  172.16.1.0/24 [110/2] via 10.1.13.2, GigabitEthernet0/1
O  10.1.23.0/30  [110/2] via 10.1.13.2 ... (and via 10.1.12.2)
R1# show ip ospf neighbor           ! two neighbours R2 and R3 in FULL
R1# show ip dhcp binding
```

Step 6: Test redundancy (backup router)

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# shutdown              ! primary link R1-R3 fails
```

Run **ping -t** or repeated pings from PC-A1 to 172.16.1.10 - after a few seconds the replies resume. **tracert 172.16.1.10** now shows 192.168.10.1 - 10.1.12.2 (R2) - 10.1.23.2 (R3) - 172.16.1.10. Then use **no shutdown** to restore the primary path. This demonstrates failover through the backup router with OSPF.

PHASE 3 - Troubleshooting scenarios

Scenario	How to create it	Diagnosis	Solution
1. Connectivity issue - PC in wrong VLAN	On SW1: *interface fa0/3*, *switchport access vlan 10* (PC-B1 moved)	PC-B1 gets a 192.168.10.x address (or none), cannot reach DEPT-B; *show vlan brief*; ping / tracert from PC	*switchport access vlan 20* on Fa0/3, renew DHCP (*ipconfig /renew*)
2. IP conflict - duplicate address	Give PC-A2 the static address 192.168.10.11 (already leased to PC-A1)	'Duplicate IP address' warning, intermittent pings; *show ip dhcp conflict*, *arp -a*	Move static hosts out of the pool: keep *ip dhcp excluded-address* range and use addresses from it for static devices; renew DHCP
3. Slow performance - congested link	Heavy FTP traffic over R1-R3 link	Monitor *show interfaces gi0/1* (load, drops)	Apply QoS (below), add bandwidth, or let OSPF load-balance

QoS example (optional - Packet Tracer supports only a basic MQC subset)

```
R1(config)# access-list 101 permit tcp any any eq 21          ! FTP / important traffic
R1(config)# class-map match-any IMPORTANT
R1(config-cmap)# match access-group 101
R1(config-cmap)# exit
R1(config)# policy-map PRIORITIZE
R1(config-pmap)# class IMPORTANT
R1(config-pmap-c)# bandwidth percent 60
R1(config-pmap-c)# exit
R1(config-pmap)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# service-policy output PRIORITIZE
```

IMPORTANT: If your Packet Tracer version rejects some QoS commands, describe QoS in the report (classification, marking, queuing) and demonstrate the other two troubleshooting scenarios. QoS is fully supported on real Cisco IOS.

Outcome and Conclusion

The network was scaled from a 2-PC office to a multi-department network with VLAN segmentation, DHCP, OSPF routing, a protected file server and a redundant path. Failures were simulated and fixed with ping, tracert and show commands. The case study shows the importance of proper design (addressing plan, VLANs), documentation and systematic troubleshooting.

Viva Questions with Answers

Q1. How do you scale a small network?

Ans: Plan addressing, segment with VLANs/subnets, use a dynamic routing protocol (OSPF), add redundancy and servers, and apply security and QoS.

Q2. What is the systematic way to troubleshoot a network problem?

Ans: Check physical layer first (cables, link lights), then data link (VLAN, trunk), network (IP, gateway, routing) and finally application, using ping, tracert and show commands (bottom-up approach).

Q3. Why do we use a backup router/path?

Ans: To keep the network available when a link or device fails; a dynamic routing protocol detects the failure and moves traffic to the alternate path.